

Phase transformations in γ based titanium aluminides

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There have been many recent advances in the development of γ based titanium aluminides for high temperature load bearing applications. Several improvements in the properties of these materials have been the direct result of better understanding of the phase transformations and microstructural evolution in this system. Engineering applications focus on the use of the lamellar microstructure, consisting of alternating plates of the γ and α_2 phases. The solidification microstructures, phase transformations in the relevant phase fields, interfacial structure, thermomechanical treatments, and alloying effects are surveyed. Some areas of current interest and unresolved issues are highlighted. IMR/359

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Introduction

The increasing demand for better fuel economy, higher operating efficiency, and reduction of pollutants in engines and turbines has prompted an intensive search for weight saving structural materials with higher operational temperature capabilities.¹⁻¹⁴ Titanium aluminide (γ) based intermetallic compounds (intermetallics) are a natural candidate for such applications because of their low density, high specific strength, high specific stiffness, high temperature strength retention, and resistance to fire. In particular, the specific modulus and specific strength as a function of temperature are much better than comparable superalloys and structural steels. The γ based titanium aluminides are a class of intermetallics that are being intensively studied as potential replacements for superalloys and steels in several high temperature structural applications in the energy, aerospace, and automobile sectors. Other promising intermetallics from the standpoint of effort expended, progress, and continuing application potential include titanium based systems (Ti_3Al and Ti_2AlNb based) and nickel based systems (Ni_3Al and NiAl based) as well as the Fe_3Al , FeAl , Nb_5Si_3 , Nb_3Al , and MoSi_2 intermetallic families.

Multiple γ based intermetallics are preferred to single phase γ because of the poor ductility and fracture toughness of the latter. However, in these two phase alloys, there is generally an inverse correlation between tensile properties and resistance to fracture, and hence microstructural control becomes important to obtain an optimum blend of properties. The γ based alloys provide a unique set of properties that can lead to substantial payoffs in a range of land based and turbine engine applications. Cast Ti-48Al-2Cr-2Nb alloys (all compositions are in atomic per

cent unless otherwise indicated) have been successfully tested in at least six components, of which two are the target of introduction efforts. The development of economic manufacturing processes and the accumulation of engineering experience is the main requirement for faster implementation of these alloys.

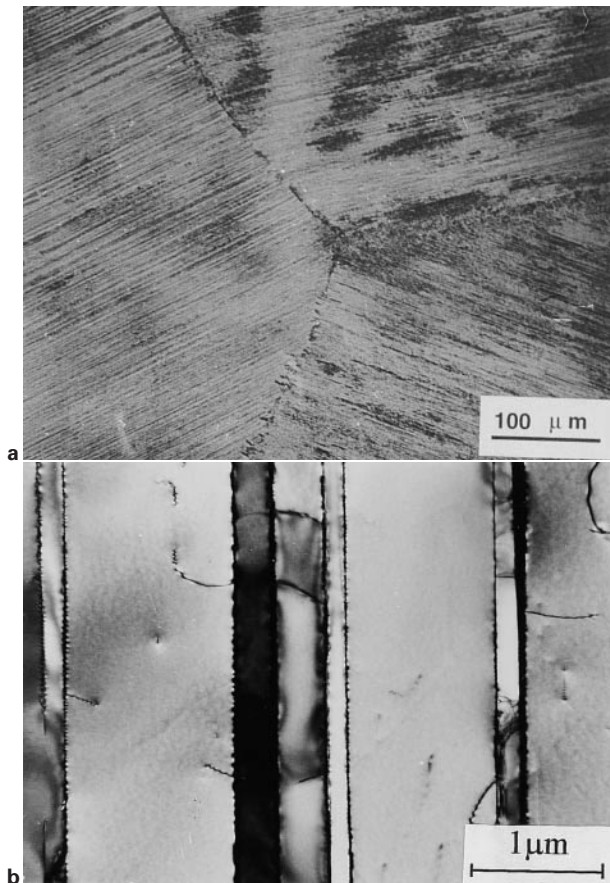
In turbine engine applications, the payoff is mostly in weight savings, since γ based alloys have half the density of superalloys and steels, and are about 10% lighter than titanium alloys.^{15,16} The most favourable properties of these intermetallics are their moduli, which approach those of superalloys at operating temperature, at half the density. Examples of components which have been tested are carbon seal supports, transition duct beams, blade dampers, compressor blades, low pressure turbine side plates, advanced engine radial diffusers, high pressure compressor blades, high speed components, and low pressure turbine blades. In the automotive sector, a castable TiAl based alloy with the composition Ti-33.5Al-1Nb-0.5Si-0.5Cr (wt-%) has been designed to increase oxidation resistance and high temperature strength.¹⁷⁻²⁰ Improvements in the properties of intermetallic automotive engine valves produced by reactive sintering have also been reported.

In binary alloys, the maximum room temperature ductility is found for the Ti-48Al composition and Cr and Nb additions are mainly made to improve the room temperature ductility and oxidation resistance respectively.²¹ This, a Ti-48Al-2(Cr or Mn)-2Nb alloy was developed with good oxidation resistance and ease of processing. The realisation that the mechanical properties are closely linked to the microstructure prompted several investigations of the correlation between microstructure and properties.²²⁻²⁴

The microstructure of these alloys consists at room temperature of α_2 ($D0_{19}$ crystal structure) grains containing parallel plates of the γ ($L1_0$ crystal structure) phase. A small volume fraction of equiaxed γ grains is also usually present. Each α_2 grain is termed a colony (Fig. 1a), and the plates of γ and the interleaved α_2 phase, which is also partitioned into parallel plates by the γ phase, are termed lamellae (Fig. 1b). A lamellar microstructure generally exhibits superior properties to single phase γ alloys.

Phase diagrams

To understand the formation of the microstructure, consider the Ti-Al binary phase diagram (Fig. 2).²⁵ The intermetallic TiAl has a wide composition range, but primarily on the aluminium rich side of stoichiometry. The phases found at room temperature in typical alloy compositions, e.g. Ti-47Al, are the γ and α_2 phases. The β (bcc, $a = 0.3178$ nm) or B2 (ordered bcc, $a = 0.316$ nm) phase can also be stabilised by suitable alloying additions and will form in some of the ternary and higher order alloys being investigated.

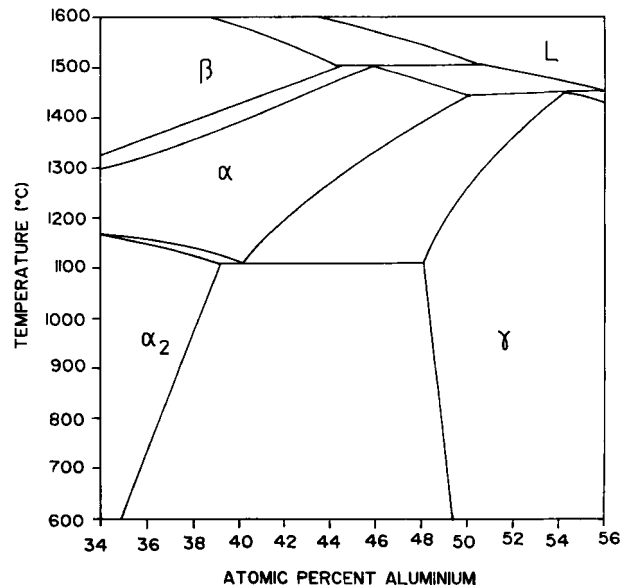


1 a Lamellar colonies in as cast Ti-48Al alloy (optical); b parallel plates of γ and α_2 phases within lamellar colony (TEM)

Important microstructures

What types of microstructures are useful? At least two types are currently being investigated. Investment casting has been used to produce components whose properties have been improved through heat treatment. Such heat treatments transform the coarse, non-uniform cast lamellar microstructure to a finer mixture of γ grains and residual lamellar regions having a grain size of approximately 100 μm . The cast alloy Ti-48Al-2Cr-2Nb with this microstructure (termed casting duplex) exhibits a reasonable balance of properties and was recently demonstrated as a viable engineering material through rigorous engineering tests. In other studies, boron was added to refine the grain structure and develop the cast XD (exothermic dispersion) alloys, such as Ti-(45–47)Al-2Mn-2Nb + 0.8 vol.-%TiB₂, which have a refined cast lamellar microstructure. Property measurements on such cast alloys show values as good or better than those of many nickel based superalloys, for which they may be substituted.

The lamellar microstructure found at room temperature can increase ductility either directly or in combination with fine recrystallised γ grains. Early alloy design efforts focused on improving ductility, and found that improved ductility occurred at about 10 vol.-% α_2 . When the α_2 phase fraction was >20 vol.-%, ductility was degraded. The ratio of lamellar colonies to equiaxed γ phase and their distribution



2 Relevant portion of Ti-Al phase diagram

can be altered to give a reasonable combination of properties with good high temperature creep resistance, acceptable tensile strength, and useable ductility.

Thermomechanical processing (TMP) has been shown to be a better method of changing the microstructure than heat treatment alone.^{26,27} Refining of the microstructure occurred when processing was conducted such that both the α and γ phases were recrystallised in the $\alpha + \gamma$ phase field. Grain morphology was found to vary considerably depending on composition, solution treatment temperature and time, cooling rate, and aging temperature and time. Recently, it has been found that a relatively small grain size (<350 μm), developing the fully lamellar microstructure with controlled lamellar spacing, and perhaps tailoring the grain boundary morphology is desirable. It is therefore necessary to use TMP methods, which first homogenise the structure, producing a fine recrystallised grain structure, then, either during hot working or after it, provide control of the grain size and lamellar spacing. The methods used are:

- (i) TMP lamellar (TMPL) materials in which the fully lamellar structure is formed during hot working
- (ii) thermomechanically treated (TMTL) microstructures in which selected alloy compositions, containing boron as a grain growth inhibitor, are heat treated to the fully lamellar form after hot working
- (iii) refined fully lamellar (RFL) structures for which both the grain size and lamellar spacing are controlled, with or without grain growth inhibitors, through post-hot working heat treatments.

It is thus possible to use TMP to alter microstructure. However, in practice, several issues have to be resolved. For example, in the α phase field, rapid α phase grain growth takes place, which decreases hot workability. At low temperatures within the $\alpha + \gamma$ phase field, heterogeneous coarsening of the intergranular γ phase occurs. Processing just below the α transus temperature (T_α) can produce a more homo-

geneous microstructure by dissolving the γ phase; however, dendritic refractory element chemical segregation persists even after multiple hot working steps. Typically, the homogenisation step is carried out before placing the ingot into any can material that may be required for hot working.

For three phase $\alpha + \beta + \gamma$ alloys, the homogenisation temperature may be extended to include the $\alpha + \beta$ phase field; however, alloy chemistries are not yet optimised to take advantage of these reactions, and too much β phase can lead to other detrimental effects.

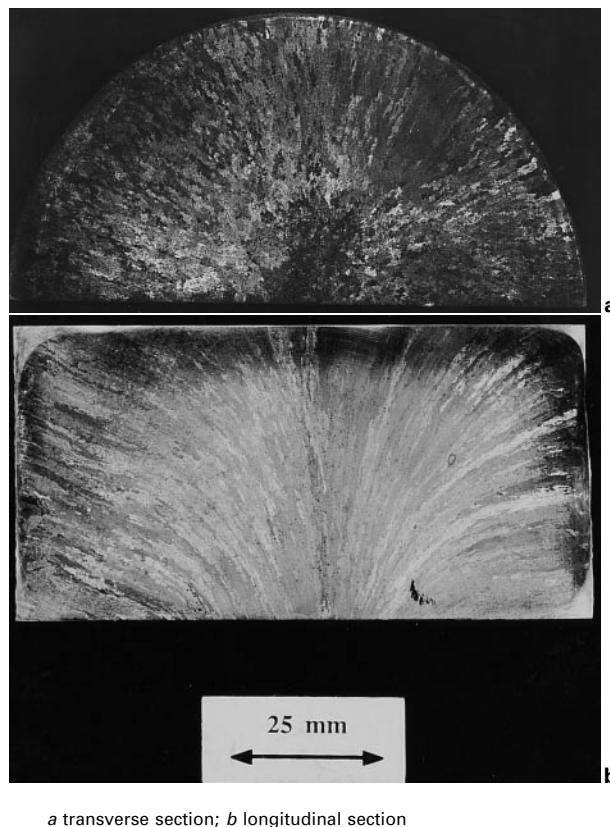
The key to further improvement of the properties is through innovative heat treatment and hot working.^{28–30} During these processes, a number of phase transformations occur to alter the colony size and grain boundary morphology, interlamellar spacing, and volume fraction and distribution of the phases. In the following, the phase transformations that can occur in these alloys will be described and important unresolved issues highlighted. The structures produced by casting will be examined, followed by a description of phase transformations on cooling from the α phase field. Phase transformations in the $\alpha + \gamma$ phase field are then described. Thermomechanical treatments are delineated. The microstructural stability under conditions of high temperature and stress, as well as dynamic precipitation, will be examined. The effect of alloying elements on the phase transformations and mechanical properties will be outlined.

Solidification structures and homogenisation

Consider the common case of alloys produced by the casting route.^{31–33} For alloy compositions in the range Ti–46Al to Ti–49Al, cooling from the liquid (L) phase follows the sequence: $L \rightarrow \beta + L$; $L + \beta \rightarrow \alpha$; $\alpha \rightarrow \alpha + \gamma$; $\alpha \rightarrow \alpha_2 + \gamma$.

The first peritectic reaction, $L + \beta \rightarrow \alpha$, was often masked by the complete transformation of the β phase to α , and this resulted in earlier erroneous interpretation of the two successive peritectic reactions as a single reaction. High temperature X-ray diffraction and solidification experiments have now shown that there are two successive peritectic reactions.^{34–36} This result is also consistent with experimental observations of directionally solidified Ti–52Al alloys.²⁴ Such structures, consisting of alternating γ and lamellar regions along the length of the crystal, are consistent with the existence of the $L + \alpha \rightarrow \gamma$ reaction. Fully lamellar microstructures can also be produced by directional solidification. These single crystal-like structures are known as polysynthetically twinned (PST) crystals.¹¹ Such crystals show remarkable properties, e.g. a Ti–48·75Al–0·05B PST crystal showed a room temperature elongation of 27·5% in the ‘soft’ direction.³¹ Mechanical testing of such crystals has provided useful information on the plastic anisotropy of fully lamellar microstructures.

Consider the structures formed by plasma arc melting of Ti–48Al, Ti–48Al–2Mn–2Nb, and Ti–52Al alloys in the form of ingots, powders, and centrifugal spray deposited tubes.^{27,37} The plasma arc melting

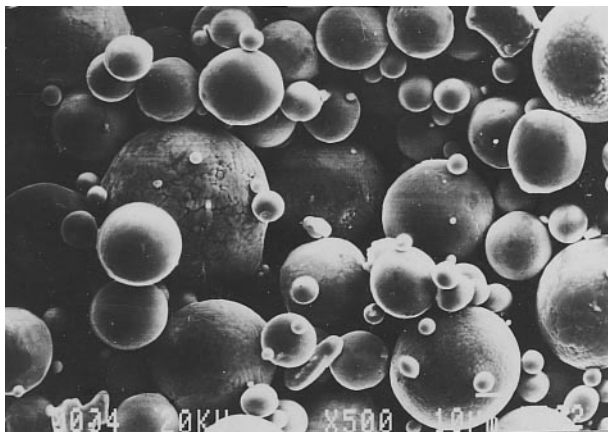


a transverse section; b longitudinal section

3 Section of Ti–48Al ingot prepared by plasma arc melting

route has been used to produce cylindrical ingots, and in a modified form to produce centrifugally spray deposited tubes and powders. Cylindrical ingots 100 mm in diameter and up to 750 mm in length have been prepared. The top transverse section showed a chill zone at the outer edge, long columnar grains extending from the edge to the centre, and apparently equiaxed grains in the central region (Fig. 3a). The middle transverse section was similar to the top region with a smaller chill zone and a smaller equiaxed grain zone. The bottom transverse section showed a much smaller chill zone and columnar region and a much larger equiaxed central region. The longitudinal section (Fig. 3b) consisted of columnar grains formed from two competing heat extraction sites, one from the side walls and the other from the bottom of the withdrawal crucible. The relative importance of the latter heat extraction increases from the top to the bottom of the ingot. The elliptical melt pool is therefore shallow at the top and deep at the bottom of the ingot. The growth of grains is along the direction of maximum thermal gradient. In the longitudinal section, this direction was given by the locus of perpendiculars erected to the tangent at each point of the melt pool/solid interface. A deep melt pool can therefore be shown to increase the volume fraction of columnar grains, which are oriented roughly parallel to the solidification direction, and this volume fraction increases from the top to the bottom of the ingot. In the transverse section, such parallel grains appear equiaxed as a result of sectioning effects.

In the case of powders produced by gas atomisation the volume fraction of the phases was a function of



4 Clean melted and gas atomised Ti-52Al powder (SEM)

alloy composition and powder size (Fig. 4). In Ti-48Al and Ti-48Al-2Mn-2Nb alloys, of powder diameters in the range 63–90 μm , the structure consisted of $\alpha_2 + \gamma$ and α_2 phases respectively. However, in the size range 500–710 μm the structure was predominantly γ . In Ti-52Al, in both size ranges the structure was mainly γ . Thus the fraction of α_2 increased with increasing cooling rate and titanium content. Observation of the microstructure of all three alloy compositions by SEM showed a dendritic structure, with a dendrite arm spacing much less than that found in ingots of Ti-52Al. The dendrites and interdendritic region were found to be aluminium lean and aluminium rich respectively.

In tubes prepared by centrifugal spray deposition (Fig. 5), the as sprayed structure in Ti-48Al and Ti-48Al-2Mn-2Nb alloys was predominantly lamellar, with a lamellar colony size in the range 5–50 μm . In Ti-52Al alloys, the structure was completely dendritic, with a dendrite arm spacing between that of ingot and powder, indicating an intermediate cooling rate.

It was, therefore, possible to vary the structure by the appropriate processing route; however, the central challenge in the case of ingots is the production of homogeneous isotropic structures without a dendritic morphology. The current practice of extensive hot working to remove the segregation and the anisotropic microstructure can then be eliminated. Hot isostatic pressing at about 1200°C is often carried out for this purpose, resulting in the formation of a duplex microstructure composed of lamellar colonies and γ grains. Such a structure has the highest ductility and lowest fracture toughness. One proposal to reduce the grain size is to choose the composition such that the β phase or the B2 phase can act as a grain refiner by reducing the growth of α grains. Another idea is to add boron, which also reduces the α grain size. In the case of powder based processing routes, the issue of a cheap process which has acceptably low contamination levels is an important one.

Crystallography and mechanisms of formation

The crystal structure of the γ phase is face centred tetragonal, $L1_0$, space group $P4/mmm$, with



5 Tube of Ti-48Al-2Mn-2Nb alloy produced by centrifugal spray deposition

lattice parameters varying with composition in the ranges $a = 0.4015\text{--}0.3975\text{ nm}$, $c = 0.4062\text{--}0.4097\text{ nm}$ (Fig. 6a). The α_2 phase has an ordered hexagonal $D0_{19}$ structure, space group $P6_3/mmc$, with lattice parameter varying with composition in the ranges³⁸ $a = 0.5783\text{--}0.5640\text{ nm}$, $c = 0.476\text{--}0.4619\text{ nm}$ (Fig. 6b).

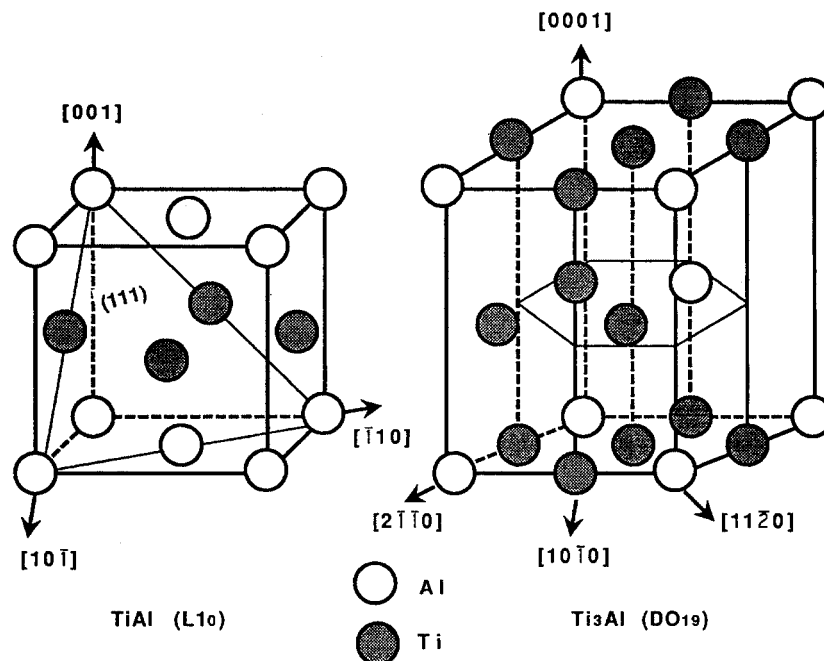
Crystallography of γ phase formation

The transformation of a hcp based structure (α) to a fcc based structure (γ) on cooling, is rare in metallic systems, whereas the reverse (fcc $\rightarrow$ hcp) transformation is common, e.g. in cobalt and Al-Ag alloys, for which stacking faults and helical dislocations can act as nucleation sites. In non-metallic systems, however, similar processes are seen in the hexagonal to tetragonal transformations in alloys of MoSi_2 and in the hexagonal to cubic transformations in Laves phases.³⁹

The Shoji-Nishiyama orientation relationship is followed in the precipitation of the γ phase in either the α or α_2 phase, i.e. $\{111\}_\gamma \parallel \{0001\}_{\alpha_2}$, $\langle 1\bar{1}0 \rangle_\gamma \parallel \langle 11\bar{2}0 \rangle_{\alpha_2}$. The close packed planes and directions are therefore parallel and since the $\{0001\}$ planes of the α_2 phase show no multiplicity, the precipitating γ plates are also parallel to one another, resulting in a lamellar structure. The γ lamellae are often twinned. Twinning is of the $(111)[11\bar{2}]$ type, not of the $(111)[\bar{2}11]$ or $(111)[1\bar{2}1]$ type. There are six interfaces that can arise from the combination of the three possible directions of the c axis of the γ phase and the ordered nature of the two phases.

Crystallography of nucleation and growth

The nucleation of the γ phase has been studied in alloys containing <45%Al, because of experimental difficulties in arresting the $\alpha \rightarrow \gamma$ transformation at the nucleation stage in alloys with higher Al levels. When such attempts are made, say by fast cooling from the α phase field, other types of transformations intervene. The nucleation of γ from the α_2 phase proceeds from stacking faults by the movement of $\frac{1}{3}\langle 1010 \rangle$ partials.⁴⁰ The growth of the γ phase has been shown by TEM



6 Crystal structure and unit cell of γ phase (left) and α_2 phase (right)

to proceed by the ledge mechanism, as in the case of analogous fcc $\rightarrow$ hcp transformations.^{41–43} The growth results in a partial shape change or distortion because an unequal number of the three possible Shockley partials are involved (Fig. 7). The ratio of 30° ($(a/6)[2\ 1\ 1]$ or $(a/6)[1\ 2\ 1]$) Shockley partials to 90° ($(a/6)[1\ 1\ 2]$) partials was 3:1. A ratio of 2:1 would have resulted in no shape change during growth, as in the case of the precipitation of hcp Ag_2Al precipitates in a fcc aluminium rich matrix. The lengthening kinetics were extremely fast and most of the changes, as a function of time, occur in the thickness of the γ plates.

The mechanism of formation of the γ phase from the α phase can conceptually be divided into the crystal structure change from hcp to fcc type, the composition change, and ordering.⁴⁴ It is difficult to distinguish these three parts, because the disordered γ phase has never been observed in solid–solid phase transformations. Changes in composition have been observed during aging, hence attainment of the equilibrium composition is the slowest step of the three outlined above.

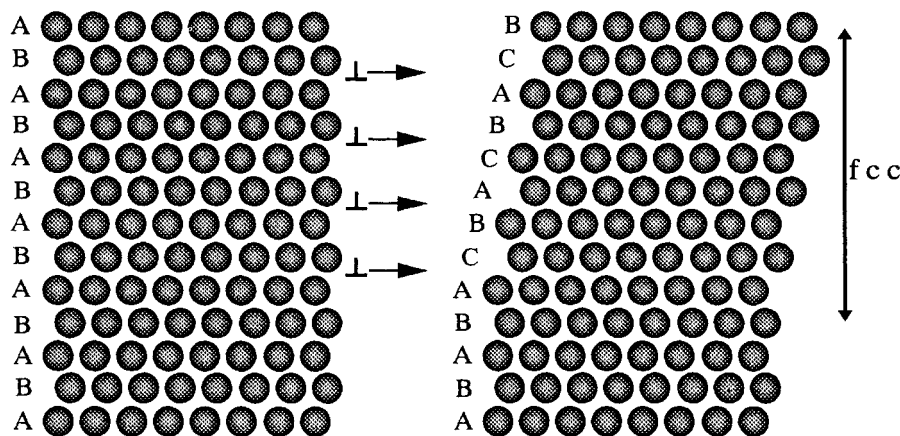
Crystallography and strain energy effects

The mechanism for the transformation of γ from α_2 has been studied, combining crystallography and strain energy studies.³⁹ The $\alpha_2 \rightarrow \gamma$ transformation differs from hcp $\rightarrow$ fcc transformations because γ lacks a triad along $\langle 111 \rangle$, creating orientation variants between adjacent γ lamellae. It is not known if these variants are formed from the shearing or the ordering process, e.g. the three 120° rotational variants may be derived either from the individual $\frac{1}{3}\langle 10\bar{1}0 \rangle$ shearing partials or from the unique orientation of the 'c' axis during ordering. Differences between $\alpha_2 \rightarrow \gamma$ and $\alpha \rightarrow$ disordered γ also arise because the driving force differs owing to the additional contributions from

free energy terms related to the ordering process and diffusion dominated effects required to obtain the equilibrium composition.

It was suggested that nucleation of the γ phase was initiated by the repeated motion of the same $\frac{1}{3}\langle 10\bar{1}0 \rangle$ shearing partial on alternate planes. While the partial in a given set of overlapping faults is the same, the partials are different in the next stack of overlapping faults. In a given stack of overlapping faults, the γ lamellae were assumed to be twinned somewhat periodically on the scale of a few angstroms. This microtwinning process is a viable mechanism for lowering the shear strain energy of the system. In particular, it provides relief in the absence of the simultaneous operation of all three coplanar shearing partials, a process that creates antiphase boundaries (APBs) in γ . Mechanistic differences between the $\alpha_2 \rightarrow \gamma$ and $\alpha \rightarrow \gamma$ transformations were attributed to structural and stoichiometric effects. Conventional TEM did show a succession of faults nucleated on a series of basal planes spread over about 100 nm thickness. The ratio of lateral spacing on the basal plane to the height affected was ~ 1 , indicating a spherical nucleus. High resolution electron microscopy showed that the thin γ plates can twin on the scale of a few angstroms. However, besides these isolated observations,³⁹ very little experimental work exists to support the microtwinning mechanism.

The hcp $\rightarrow$ fcc transformation can be achieved by the motion of $\frac{1}{3}\langle 10\bar{1}0 \rangle$ partials on alternate basal planes of the hcp phase (Fig. 8). One mechanism (mechanism A) assumes the simultaneous operation of all three coplanar shearing partials R_1 , R_2 , and R_3 on alternate layers. From the observations of twins and of lamellae < 6 planes thick, it was decided that this mechanism was not operating in the $\alpha_2 \rightarrow \gamma$ transformation.³⁹ Another mechanism (mechanism B) assumes the motion of a single partial R_1 gliding on alternate planes of the hexagonal crystal. All the shear



7 Movement of Shockley partials producing hcp $\rightarrow$ fcc structure change: passage of partials on alternate $(0001)_{\text{hcp}}$ planes produces ABCABC ... stacking sequence

will be in the same direction and the volume of sheared material increases with the motion of every partial. Mechanism B is favoured in simple metals. In mechanism C, the shear stress developed by mechanism B can be relieved by the reverse motion of the same partial. The partials R_1 and $-R_1$ glide on alternate planes, but at the twin boundary two adjacent planes are sheared. Twins which are $2n$ layers thick are produced and stress relief is achieved if the cumulative thickness of γ and γ_{twin} variants approximately match. In the $\alpha_2 \rightarrow \gamma$ transformation this R_1 , $-R_1$ mechanism (mechanism C) should be favoured because in the $L1_0$ structure, the vectors $R_2 - R_1$ and $R_3 - R_1$ are not necessarily lattice translations, they may be superpartial vectors. If, as seems likely, R_1 is the twinning Shockley partial of the $L1_0$ structure, then both $R_2 - R_1$ and $R_3 - R_1$ dislocations would leave behind APBs on planes. The fact that mechanism C operates means that twin interfaces between γ and γ_{twin} are energetically favoured over the creation of two APBs every six planes.

Twinning in γ phase

Twinning is an important mode of deformation in superlattices with the $L1_0$ structure.^{45–50} Detailed TEM based investigations of twinning in γ based alloys has been carried out.^{51–57} Several investigations of the CuAuI superlattice, which also has the $L1_0$ structure, have been carried out and $\{111\}$ twinning was determined to be an important deformation mechanism. Twins are usually classified as type I (K_1 and η_2 are rational planes and a rational direction of the parent lattice), type II (K_2 and η_1 are rational) or compound (all four crystallographic elements are rational).⁵⁸ Here, K_1 is the invariant plane of twinning shear (twinning or composition plane), K_2 the undistorted (but rotated) plane (conjugate or twinning plane), η_1 is the direction of shear, and η_2 is the conjugate shear direction (Fig. 9a).

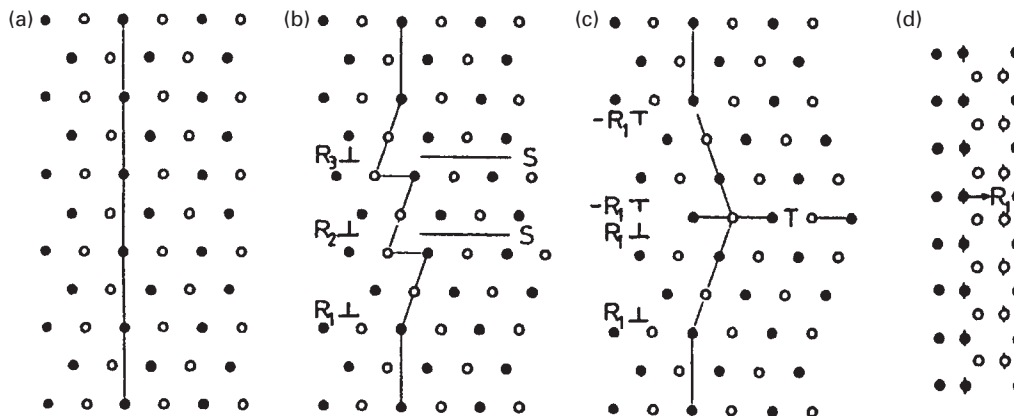
Plastic deformation of γ by mechanical twinning is associated with $\frac{1}{6}\langle 112 \rangle \{111\}$ shear. Only $\frac{1}{6}[11\bar{2}](111)$ twinning does not alter the $L1_0$ structure. Shear in the reverse direction, along $-\frac{1}{3}[11\bar{2}](111)$, is the anti-twinning mode. The other partials, $\frac{1}{6}[211]$ and $\frac{1}{6}[1\bar{2}1]$, lead to pseudo-twinning. True twinning was

prevalent in γ alloys. The occurrence of anti-twinning and the related formation of the $9R$ structure remains unresolved (Fig. 9b and c). In an early study of twinning in TiAl, the specific variants of the $\langle 112 \rangle \{111\}$ cubic twin which formed during deformation were identified and shown to correspond to the true modes of the superlattice. The shear magnitude was also measured.⁴⁵

The heterogeneous nucleation of twins in γ based alloys has been related to: the dissociation of superdislocations, the expansion of stacking fault loops resulting from the dissociation giving rise to twin nucleation; microtwins nucleated by the superposition of extended superlattice extrinsic stacking faults on alternate $\{111\}$ planes; a pole mechanism; and interfacial boundaries.¹ Twins appear to grow by the movement of ledges, which have the character of Shockley partials. Occasionally twins appear to nucleate from grain interiors, but nucleation on individual misfit dislocations of a semicoherent γ/γ boundary has also been observed.^{1,58}

In two phase $\gamma + \alpha_2$ structures, twins appeared to grow by successive formation and bowing out of twinning dislocations from grain boundary regions to adjacent lattice planes of the γ matrix.^{59,60} The leading dislocations moved slowly and came to a halt in the interior of the parent grain. Once formed, the next dislocation moved forward relatively rapidly to the vicinity of the lead dislocation and apparently pushed it a little further into the grain, before itself coming to a halt at a very short distance behind its leader. Repetition of this event then produced a microtwin.

High resolution electron microscopy has been used to verify the fully coherent nature and high state of perfection of the K_1 interface.⁵⁸ Lattice images from twin/twin and twin/matrix interfaces have yielded information on the magnitude of the total shear across twins. This has been measured and shown to correspond to one twinning dislocation having moved through each lattice plane, thus indicating the perfection of the twin structure. A parallel set of twinning dislocations of the same sign typifies a tapering twin. The distance between successive twinning dislocations initially increased with distance from the tip in a manner similar to that expected from a dislocation



a schematic structure of parent α_2 phase; b modified α_2 structure transformed to $L1_0$ by three Shockley partials with $b = R_1, R_2$ and R_3 : note slip steps and APBs on planes S; c six layers of α_2 phase transformed to $L1_0$ by dislocations with $b = \pm R_1$: note twin plane T; d two close packed planes of either modified $D0_{19}$ or $L1_0$ structure: plain atoms are on A layer and ticked atoms on B layer; shear which retains nearest neighbour bonding occurs only along Shockley vector R_1

8 Mechanism of creating γ phase from α_2

pile-up and then remained approximately constant when the number of twinning dislocations was more than 15. Such data are useful in determining the shape of the twin tip.

In situ deformation experiments in the TEM have verified that twins can form by the motion of twinning dislocations (Shockley partials in this case).⁵⁸ The twin velocities were small and roughly constant ($\sim 50 \text{ nm s}^{-1}$) and the separation of the twinning dislocations was constant during growth. Backward motion was observed on release of the applied stress. Thus the growth velocity was governed by a friction force, which acts equally on all twinning dislocations. Nucleation results from the dissociation of a superdislocation and subsequent interaction of the twinning dislocations to form a pole source. This distribution of twinning dislocations as well as theoretical calculations support the lenticular model of an enclosed twin and the pile-up model of a twin tip. Twin-twin interaction has also been studied in considerable detail and an example is shown in Fig. 10.⁶¹

Transformations from α phase field

Morphologies

The transformation from the α to the γ phase can occur on cooling from the α single phase field or on cooling from the $\alpha + \gamma$ phase field. Considering the first case, slow cooling produces the lamellar structure discussed above. However, faster cooling rates produce a dramatic change in the morphology, with the formation of a variety of non-lamellar microstructures such as Widmanstätten plates, mottled structures, 'feathery' regions, and massive and martensitic decomposition products^{62–65} (Fig. 11). In Ti-(45–50)Al alloys, the Widmanstätten structure formed at intermediate cooling rates in the high aluminium alloys.⁶⁶ For given cooling rate, Widmanstätten structure volume fraction increased with increasing aluminium content. Patchy γ grains were also often found in this structure.⁶⁷

The massive transformation occurs in Cu–Zn, Cu–Al, Cu–Ga, Fe–Ni, silver based, and titanium

based alloys.⁶⁸ In an early study of quenched Ti–48Al alloys, the following morphologies were observed:⁶⁷

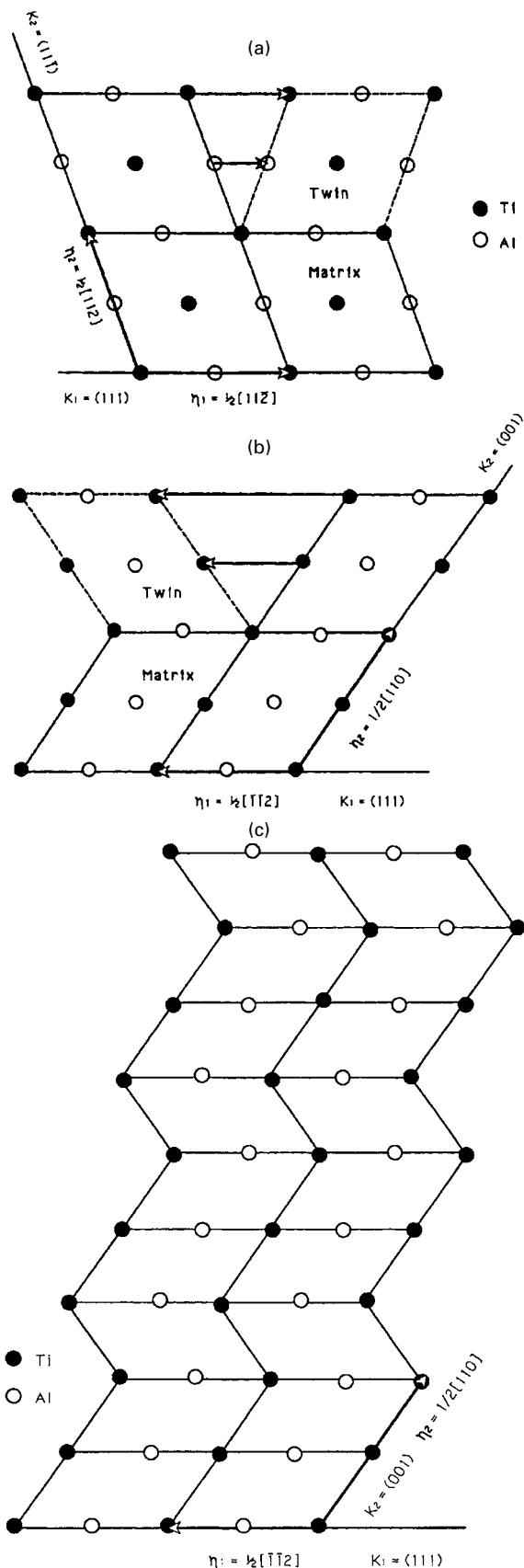
- (i) furnace cooling (FC): lamellar
- (ii) air cooling (AC): Widmanstätten structure in 3 or 4 orientations within a grain, mottled regions, and feathery structures
- (iii) oil quenching (OQ), water quenching (WQ), ice water quenching (IWQ): dark deeply etched patchy regions in a featureless matrix. These regions appear to have nucleated largely at α grain boundaries and then spread in a spheroidal fashion. Often these regions were found within grains at some distance from grain boundaries. The formation of these regions was attributed to the massive transformation. γ based alloys with aluminium contents $\leq 45\%$, WQ from the α phase field, transformed completely to α_2 and no massive morphology (γ_m) was formed. In a study of Ti-(43–50)Al alloys, WQ from the α phase field,⁶⁹ γ_m was formed only in the range 46.54–50%Al.

Another study suggested that the cooling rate effect on morphology can be summarised as:

- (i) IWQ: $\gamma_m^* + \gamma_f + \gamma_w + L_f$
- (ii) WQ: $\gamma_f^* + \gamma_m + \gamma_w + L_f$
- (iii) AC: $\gamma_f^* + \gamma_w + L_f$
- (iv) FC: $L_{eq}^* + DCL + CCL$

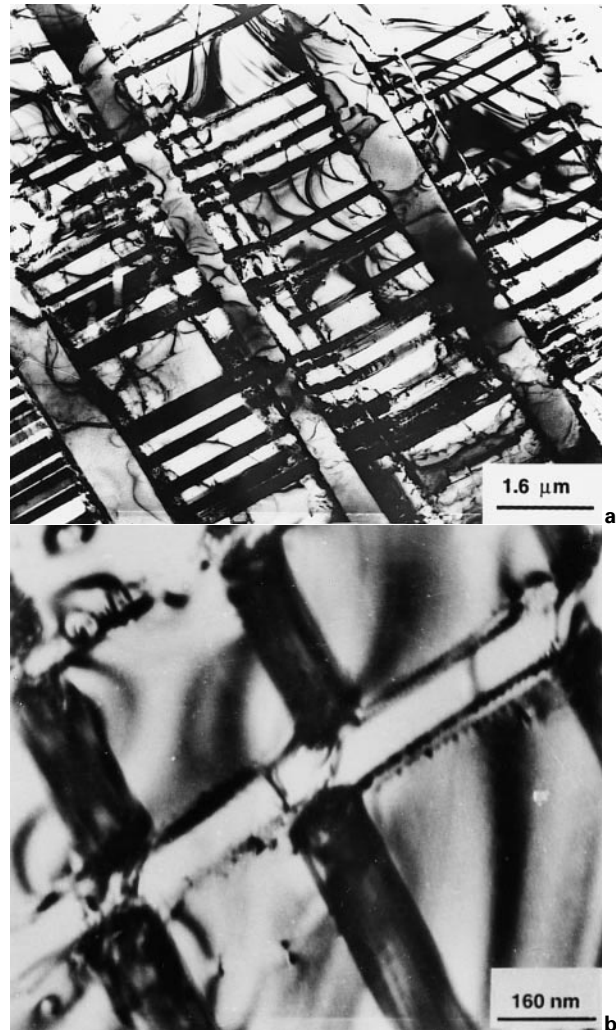
where L_{eq} represents equilibrium lamellae (1–5 μm in width), L_f fine lamellae (0.02–0.2 μm in width), γ_m massive morphology, γ_w Widmanstätten-like morphology, γ_f feathery γ needle, DCL discontinuously coarsened lamellae, CCL continuously coarsened lamellae, and * the dominant morphology.⁷⁰

In a more quantitative study, the microstructure of Ti–48Al cooled from the α phase field was shown³⁰ to vary from lamellar structure + γ grains to lamellar structure to fine lamellar structure with Widmanstätten morphology as cooling rate was increased from 0.06 to 3.6 K s^{-1} . If cooling rate exceeds 10 K s^{-1} a very fine Widmanstätten structure was formed, comprised of the γ phase produced by a



a primary twinning operation (K_1, η_1); b anti-twinning operation ($K_1, -\eta_1$); c 9R structure created by an anti-twinning operation on every third plane in twinned region

9 Structural model of twinning in γ -TiAl



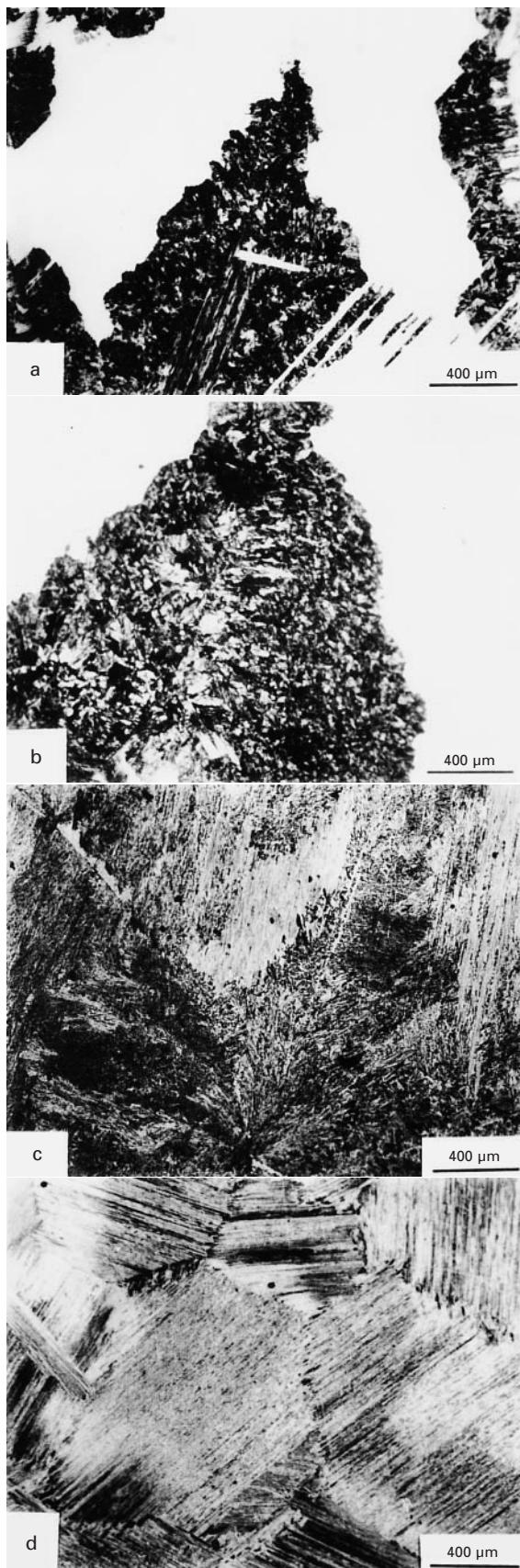
a annealing twins formed approximately perpendicular to original lamellae; b twin-twin intersection showing displacement of twins owing to intersection

10 Twinning in Ti-47Al aged at 1200°C for 1 week (TEM)

massive transformation. The defects in γ_m were recovered by tempering at low temperatures, resulting in equiaxed γ grains with finely distributed α_2 particles. Other studies are described in the relevant sections below.

Kinetics

Both time-temperature-transformation (TTT) and continuous cooling transformation (CCT) diagrams are valuable sources for understanding phase transformations and such diagrams have been painstakingly constructed by several investigators using a variety of experimental techniques.⁷¹ Using electrical resistivity techniques, a CCT diagram has been constructed for binary Ti-48Al alloys (Fig. 12a).⁷² This diagram shows the various morphologies that form as a function of cooling rate; γ_{Ls} and γ_{Lf} represent the start and finish of lamellar structure formation respectively, whereas γ_{ms} and γ_{ws} represent the start of formation of the massive and feathery + Widmanstätten morphologies respectively. The results of another investigation of the same alloy



a water quenching; b ice water quenching; c air cooling; d furnace cooling

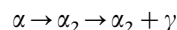
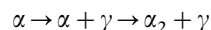
11 Ti-48Al-2Mn-2Nb alloy after heat treatment at 1420°C for 2 h, followed by cooling at different rates (optical)

composition,³⁰ but in a different range of cooling rates, is shown in Fig. 12b, which indicates the cooling rate and times required to form the γ grain, fully lamellar, Widmanstätten and massive morphologies. A TTT-CCT diagram for a β phase containing alloy of composition Ti-46.5Al-2Cr-2Nb-0.9Mo-0.2B is shown in Fig. 12c.²⁶ This diagram describes the formation of lamellar, cellular, acicular, and massive morphologies, the massive morphologies occurring at the fastest cooling rates. Starting from the lamellar structure, a TTT diagram has been constructed for a Ti-45Al-2Cr-2Nb alloy showing the transformation of 5, 50, and 95% equiaxed morphology (Fig. 12d).⁷³ Schematic CCT diagrams for the alloys Ti-44Al-8Nb and Ti-44Al-4Ta-4Zr-0.2Si, depicting the decomposition of the β phase when the alloy is cooled from the $\alpha + \beta$ phase field, have also been constructed.⁷⁴ Various phase transformations, including discontinuous coarsening, the precipitation of γ from β , and the formation of the ω phase, were observed in these alloys (Fig. 12e).

Lamellar microstructure

Formation mechanisms

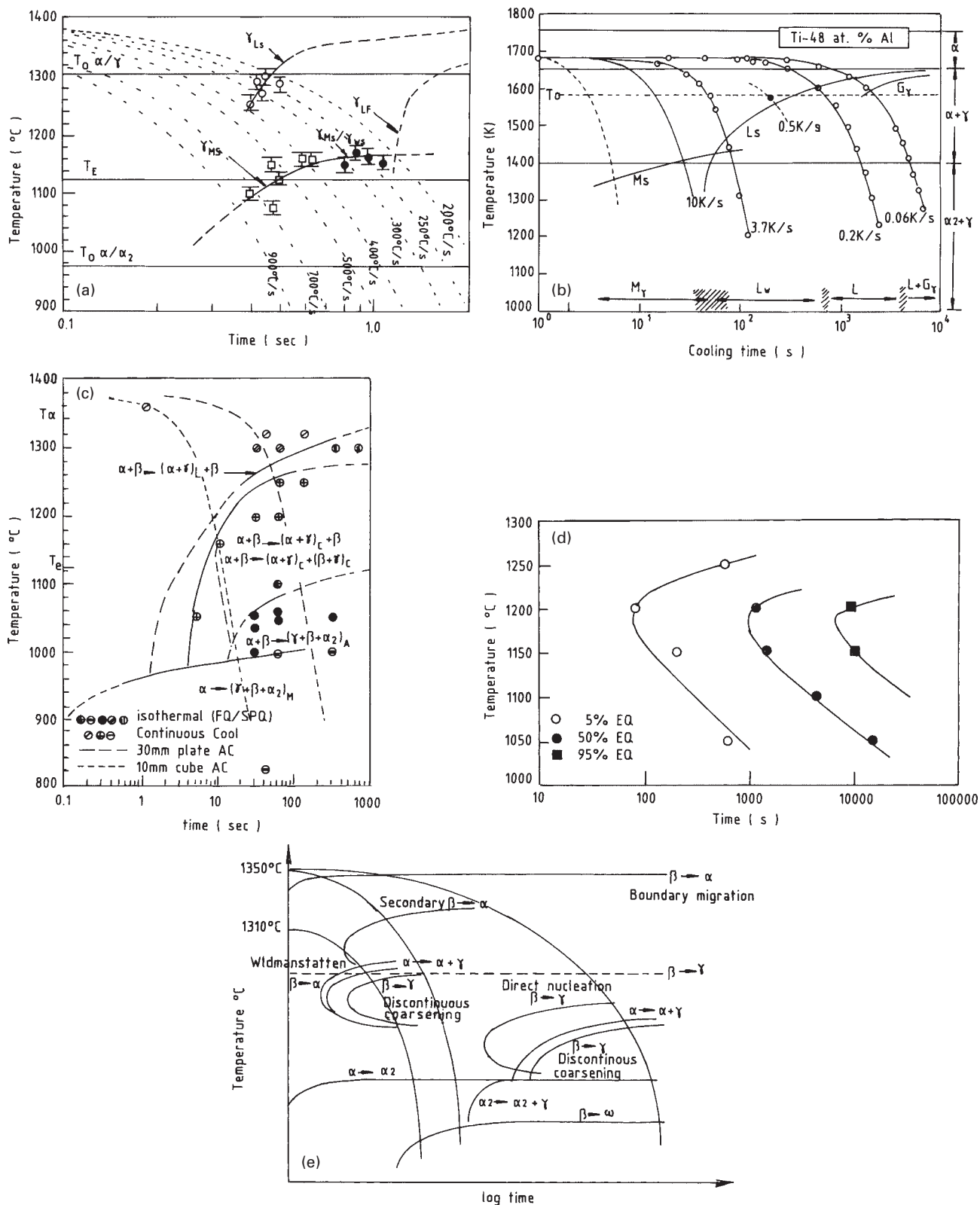
The lamellar morphology can be formed by either of the sequences



i.e. the ordering of α to α_2 can follow or precede the structural transformation from α to γ . Broadly, the first sequence is followed by alloys with Al contents >45%, whereas compositions with lower aluminium contents follow the second scheme. Experimental measurements of T_0 , the temperature at which the free energies of the α and α_2 phases are equal, as a function of composition for the $\alpha \rightarrow \alpha_2$ reaction have been carried out.^{75,76} A simple comparison with the estimated T_0 temperature as a function of composition for the $\alpha \rightarrow \gamma$ reaction shows that the crossover of these T_0 curves occurs at a value of Ti-44Al. This is consistent with the earlier statement that the ordering occurs first in alloys with aluminium contents <45%. Experimentally, an isothermally forged Ti-47Al-1Cr-1V-2.5Nb alloy showed three types of lamellar structure:

- (i) lamellae formed in the α matrix during cooling after a high temperature heat treatment:
 $\alpha \rightarrow (\alpha/\gamma)_L \rightarrow (\alpha_2/\gamma)_L$
- (ii) lamellae formed in the α_2 matrix during rapid cooling after a lower temperature heat treatment: $\alpha \rightarrow \alpha_2 \rightarrow (\alpha_2/\gamma)_L$
- (iii) lamellae formed in the γ matrix during recrystallisation and cooling via the process
 $\gamma + \alpha \rightarrow (\gamma/\alpha)_L \rightarrow (\gamma/\alpha_2)_L$

where subscript L denotes the lamellar morphology.²⁶ The final structure consisted at room temperature of parallel plates of γ and α_2 phases, the plates of γ dividing the prior α grain. The interlamellar spacing can vary from ~10 nm to several micrometres, depending on the cooling rate. The colony size following casting can be varied by hot working/heat treatment, which changes the grain size by recrystallisation and grain growth processes.



12 **a** CCT diagram for Ti-48Al alloy based on electrical resistivity measurements;⁷² **b** CCT diagram for Ti-48Al cooled from 1410°C: M_γ , L_w , L , and G_γ refer to massive γ , Widmanstätten structures, fully lamellar structure, and γ grains;³⁰ **c** TTT-CCT start times for Ti-46.5Al-2Cr-2Nb-0.9Mo-0.2B alloy: subscripts L, C, A and M represent lamellar, cellular, acicular, and massive reactions respectively; dashed lines indicate calculated air cooling curves;²⁶ **d** TTT diagram for Ti-45Al-2Cr-2Nb alloy, showing transformation curves for 5, 50, and 95% equiaxed morphology;⁷³ **e** schematic CCT diagram for Ti-44Al-8Nb and Ti-44Al-4Ta-4Zr-0.2Si alloys, simulating decomposition of β phase while cooling from $\alpha + \beta$ phase field⁷⁴

From a TEM study⁴³ of Ti-46Al and Ti-48Al, the mechanism of formation of lamellar structure was proposed to commence with a prenucleation stage corresponding to the formation of a local fcc type stacking sequence by the movement of Shockley partials, i.e. by the dissociation of a perfect $(a/3)\langle 1120 \rangle$ dislocation into two Shockley partials, $(a/3)\langle 1120 \rangle \rightarrow (a/3)\langle 1010 \rangle + (a/3)\langle 0110 \rangle$ bordering a stacking fault. Repetition of this mechanism on alternate basal planes gives the crystal structure change. Prenucleation is followed by nucleation and growth stages involving both ordering and ledge movement. Diffusional growth takes place by the ledge mechanism. As is generally observed in such cases, the lengthening takes place much faster than thickening.

According to Denquin and Naka, the mechanism of lamellae formation implies the occurrence of various boundaries between γ domains: order domain boundaries (ODBs), antiphase boundaries (APBs), twin boundaries (TBs), and pseudo-twin boundaries (PTBs).⁷⁷ No APBs were observed in this investigation, which has been attributed to the fact that APBs can interact with other types of domain boundaries which can act as sinks. This interaction can lead to 'mixed' boundaries:

ODB + APB = MODB (mixed ODB)

TB + APB = MTB (mixed TB)

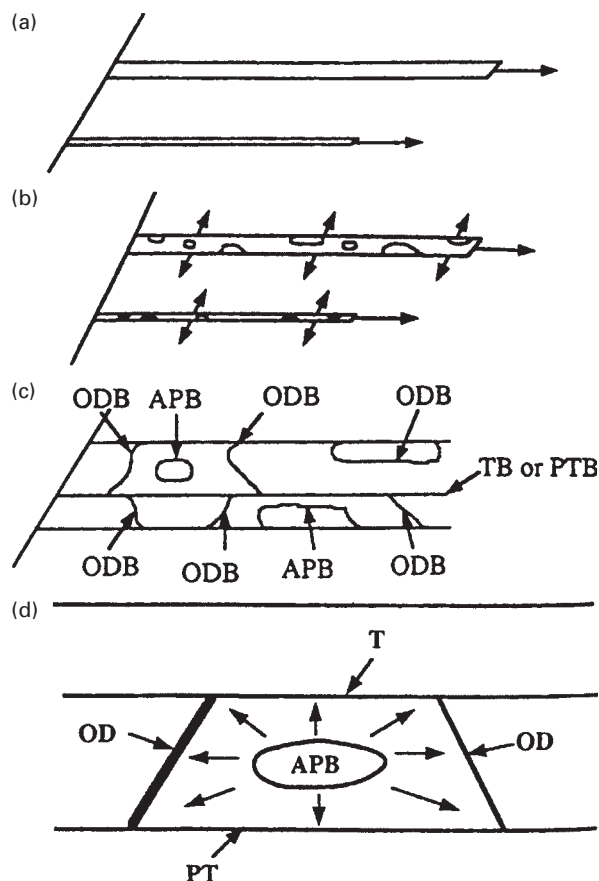
PTB + APB = MPTB (mixed PTB)

Further coupling of an APB with a MB gives a pure boundary.

It was suggested⁷⁷ the growth of the γ phase can be summarised as Shockley partial propagation (Fig. 13a). Within the fcc structure found between the partials, other domains are nucleated at independent sites, due to the formation of the $L1_0$ structure and the consequent orientation variants corresponding to the possible orientations of the 'c' axis of the $L1_0$ structure (Fig. 13b). This ordering is assumed to take place during the growth and coarsening of the γ phase and its velocity is assumed to be less than the migration of the Shockley partials. In the next stage, APBs are formed within the lamellae and ODBs are formed by order domains meeting during their growth (Fig. 13c). Twins and pseudo-twins can also be formed, leading to TBs and PTBs. Finally, the various defects can interact with each other leading to the schematic configuration shown in Fig. 13d.

Interfaces formed in lamellar microstructures

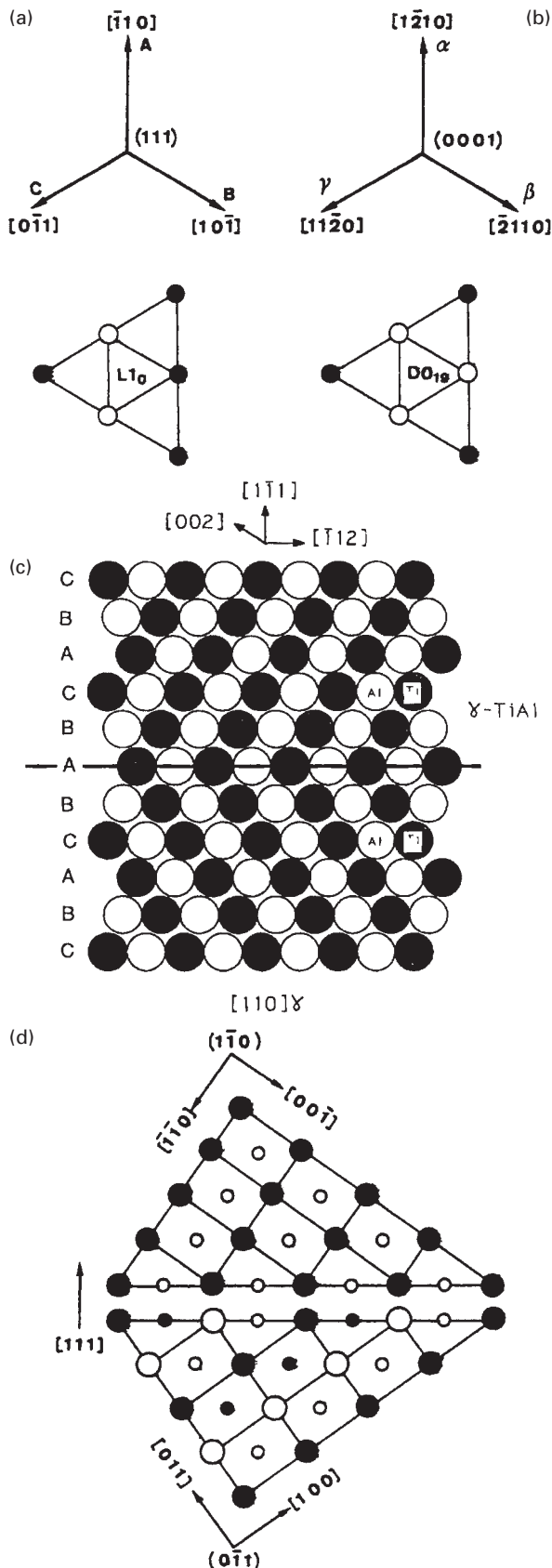
The types of interfaces that can form between the γ and twinned γ phases and between the γ and α_2 phases depends on the orientation relationship and the habit plane.^{1,78-80} The lamellar structure forms by precipitation of the γ phase from the α or α_2 matrix. The orientation of the γ variants in the individual domains can be described by multiples of 60° about $(111)_\gamma$ (Fig. 14a). In the $L1_0$ structure, on the (111) plane, the $[110]$ direction differs from the $[0\bar{1}1]$ and $[101]$ directions, while on the (0001) plane of the α_2 phase, the three $\langle 1120 \rangle$ directions are equivalent. Since there are six orientations of $[110]_\gamma$ with respect to $\langle 1120 \rangle_{\alpha_2}$, the γ phase has a domain structure of six different variants. Specifically, the



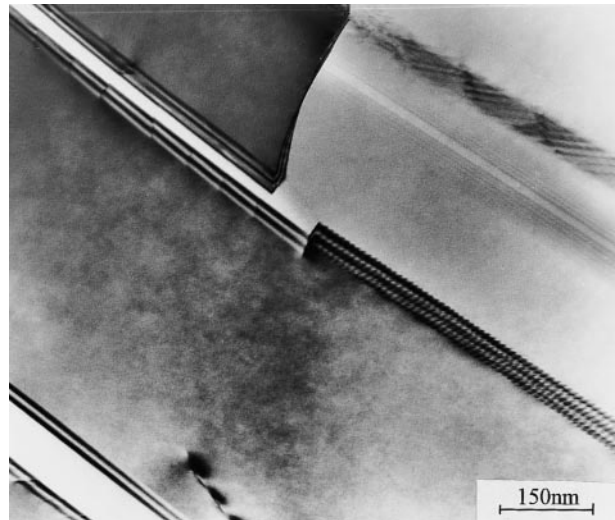
a Shockley partial propagation; b nucleation of orientation variants at independent sites: this ordering reaction is assumed to take place during lamellar growth and coarsening; c encounter of orientation variants after their growth, to form order domain (OD) boundaries as well as antiphase boundaries (APBs); lamellar coarsening can also lead to formation of twin or pseudo-twin boundaries; d representation of coupling of APB with OD, twin (T), and pseudo-twin (PT) boundaries

13 Schematic representation of possible mechanism of γ lamella formation

following orientations may occur between two neighbouring ordered domains: A parallel to A' , A parallel to B' , and A parallel to C' ; A antiparallel to A' , A antiparallel to B' , and A antiparallel to C' ; where A, B, C and A' , B' , C' represent three directions in the two domains respectively.²² In the first case, a translation order/fault interface is formed between the two neighbouring domains, otherwise no interface is produced. For the second and third cases, which are equivalent, a 120° rotation produces an order/fault interface between the two neighbouring domains, otherwise no interface is produced. The adjacent lamellae are in matrix/matrix relation if the tetragonality of the γ phase and the different atomic species are ignored. The misorientations across domain boundaries are the same as those of lamellar interfaces, but the habit plane is usually not of type $\{111\}$. In this case, the 'c' axes of two neighbouring domains are perpendicular. When A and A' are antiparallel, the adjacent lamellae match exactly: two adjacent domains have the $(111)[\bar{1}12]$ true twinning relation (Fig. 14b). For the other two antiparallel cases, which are equivalent, pseudo-twins are formed and approximate matching of close packed directions



14 *a* $\langle 110 \rangle$ direction and atomic arrangement on $\{111\}$ plane in γ ; *b* $\langle 11\bar{2}0 \rangle$ direction and atomic arrangement on basal plane of α_2 ; *c* atomic model of coherent $(111) [\bar{1}12]$ deformation twin boundary in γ phase along $\langle 110 \rangle_\gamma$ direction; *d* atomic arrangement of pseudo-twin boundary in γ phase



15 Interfacial structure observed at γ/γ interfacial boundary (TEM)

takes place (Fig. 14c). Regular alternation of the 'c' axis also occurs for the antiparallel cases.

In the case of the α_2/γ interfaces, a significant mismatch occurs owing to the differences in crystal structure and lattice parameters.^{81,82} Similarly, misfit strains are present across the 60 and 120° interfaces of γ variants as a result of their tetragonality. The unrelaxed state of such interfaces implies a misfit strain of $\sim 1\%$, which cannot be accommodated elastically. The interfaces therefore contain arrays of misfit dislocations, which decrease the misfit strain energy and improve the interfacial coherency by producing local relaxations. Processing and heat treatments can also change the nature of the interfaces.

The α_2/γ interfaces at an atomic level were mostly composed of coherent terraces parallel to $(111)_\gamma$ and arrays of regularly spaced ledges, thus the habit plane deviates from the ideal $(111)_\gamma$ (Fig. 15).⁴¹⁻⁴³ The ledges behave like Shockley partials, which implies that the growth and dissolution can be mechanistically thought of as occurring by the passage of these dislocations on alternate $(111)_\gamma$ planes. In a few cases, particularly in alloys subjected to hot working, interfaces can contain a tilt misfit component because of interfacial dislocations with an additional $(111)_\gamma$ plane parallel to the two γ phases. Significant coherency stresses are present at the interfaces and, using the curvature of dislocation loops at the interfaces, these stresses have been estimated¹ to be about 130 MPa.

It is known that the deformation behaviour is strongly dependent on the orientation of the lamellae with respect to the stress axis.²² Experimental studies have been conducted to elucidate this orientation dependent behaviour at a macroscopic and microscopic level.⁸³⁻⁸⁵ However, several issues remain unresolved, including:

- (i) it has been calculated that the extent of dissociation of an ordinary dislocation is twice as large at the α_2/γ and γ/γ interfaces than in the bulk γ phase, suggesting enhanced slip along the interfaces when the crystal is in a

soft orientation.⁸⁰ The increased dislocation width would also reduce the mobility of a $\{111\}_{\frac{1}{2}}\langle 1\bar{1}0\rangle$ screw dislocation along a true twin boundary, but increases it along all other types of interfaces. These predictions can be experimentally tested

- (ii) it is anticipated that channelled dislocation motion or supersoft deformation modes will exist. However, experimental work shows strain mostly away from the α_2/γ interfaces. Shear strain in the soft direction has been experimentally related to twinning and partly to slip in the bulk of the γ phase. This difference between prediction and experiment is unresolved
- (iii) crack propagation is predicted to be more difficult along α_2/γ and true twin boundaries, but this has not been experimentally confirmed
- (iv) the role of interfacial structure on the slip transfer has been studied in post-fracture TEM investigations. However, *in situ* deformation work is needed to understand the interaction between interfacial dislocations, ledges, and incoming dislocations from the matrix.

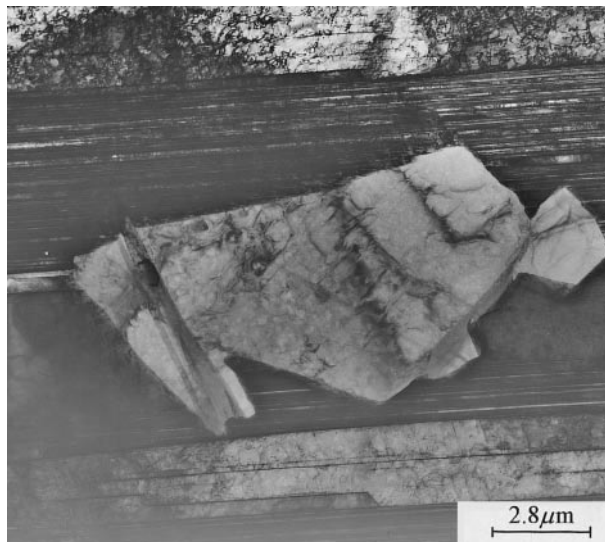
Such investigations will require: *in situ* straining in the TEM; monitoring of the dynamics of slip transfer across interfaces; use of monotonic and cyclic loading to differentiate between mechanisms arising from constant sign of stress and those prevailing when both compressive and tensile loading occurs; and determination of the kinetics of crack propagation and relative ease of crack movement across interfaces.

Massive transformations

Morphology and mechanisms

Investigations using electron probe microanalysis revealed that several of the structures formed on fast cooling did so in a composition invariant manner. Such composition invariant transformations differ from conventional massive or martensitic mechanisms since the parent and product phases differ in the degree of order, the α phase being disordered, while the product phase is ordered. However, the morphologies are conventionally described as massive or martensitic. The formation mechanisms of these structures has been the subject of a number of investigations, mainly using electron microscopy, since optical microscopy cannot provide sufficient detail. To distinguish between massive and martensitic products is also important, since both types of transformation are common in conventional titanium alloys, e.g. TEM studies of a Ti-48Al-2Mo alloy showed a high density of microtwins and other defects which were associated with the martensite-like structure of water quenched samples.⁸⁶

A Ti-48Al-2Mn-2Nb alloy was heat treated at 1420°C, varying initial microstructure, isothermal reaction time, and cooling rate.⁶² The effect of increasing the cooling rate from the α phase field was to change the structure from lamellar on furnace cooling to a complex structure on water quenching. This structure consisted of grain boundary allotriomorphs, a mottled structure, a fine lamellar structure, and dark patchy regions. Martensitic morphologies such

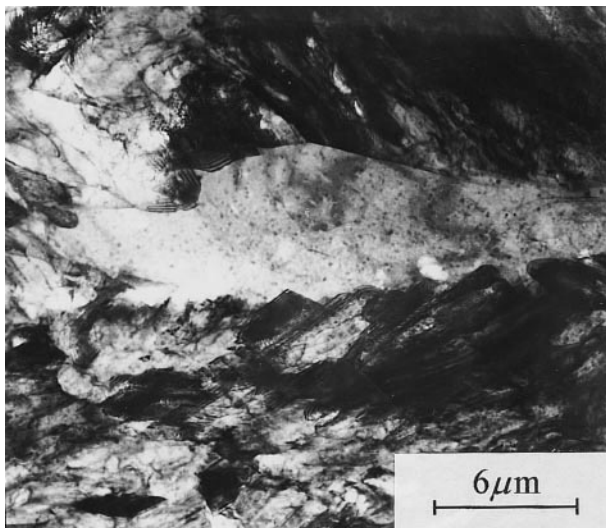


16 Coexistence of fine lamellae and equiaxed grains in Ti-48Al-2Mn-2Nb alloy after heat treatment at 1200°C for 1 week followed by heating to 1420°C for 1 h (TEM)

as acicular and fine lamellae (Fig. 16) were observed and the coexistence of massive and martensitic mechanisms was also found to be plausible.

A Ti-48Al alloy, ice water quenched after heat treatment at 1683 K for 4 h, showed that γ_m was nucleated at the original α grain boundaries and propagated into the grain.⁸⁷ If the grain size was small, the $\alpha \rightarrow \gamma_m$ transformation would be complete. For large grain size, untransformed α remained in the centre of the grain, which transformed to fine lamellae.⁸⁸ The $\alpha \rightarrow \gamma$ mechanism was interpreted in terms of a shear-like motion of atoms. In Ti-48Al alloys water quenched from the α phase field, a fine lamellar structure (a few nanometres thick) coexisted with the massive product, whereas in Ti-(40–47)Al, the metastable α_2 single phase was frozen in. In Ti-48Al, HREM showed that the antiphase domain boundaries (APDB) were continuous in some places, implying that the γ plates grow by the migration of partials after the $\alpha \rightarrow \alpha_2$ ordering reaction. Similarly, observations of APD size produced by a direct quench aging experiment was used to determine T_0 for $\alpha \rightarrow \alpha_2$ ordering.⁷⁵ In Ti-43Al, ordering was shown to take place first, followed by APD growth before γ lamellae precipitation. Transmission electron microscopy was also used to show, in Ti-40Al, that the formation of differently oriented regions coexisting within the same γ lamella was closely related to the APBs in the α_2 matrix.⁸⁹

The massive transformation was observed by AEM in Ti-49Al, Ti-48Al-2Mn-2Nb, Ti-55Al-25Ta, and Ti-50Al-20Ta alloys.⁹⁰ Issues such as the mechanism of nucleation and orientation relationship with nucleating grain, orientation relationship and nature of the boundary between γ_m and grain being consumed, whether there is an intermediate fcc phase, and how the structural defects develop were addressed. Conventional solution treatment and quenching experiments and electron beam melting (to achieve faster cooling) were used. It was proposed that γ



17 As cast Ti-48Al-2Mn-2Nb alloy after heat treatment at 1420°C for 1 h followed by water quenching, showing massive region with faceted interface and several defects associated with massive region and surrounding matrix (TEM)

developed first from grain boundaries as lamellae in one of the grains and these lamellae intersected and spread into the adjacent grain in a massive manner. Therefore there was no orientation relationship between γ_m and the grain being consumed, while the expected orientation relationship existed between γ_m and the first grain, which was inherited from the lamellae. γ_m grew as a fcc phase after initially growing with the $L1_0$ structure. The massive fcc phase then ordered to the $L1_0$ structure, producing APDB-like defects. The advancing γ_m interface tended to facet parallel either to one of its four $\{111\}$ planes or to the basal plane in the grain being consumed (Fig. 17). Thin microtwins and α_2 platelets then formed in γ_m , a transformation that may be driven by transformation stresses as well as supersaturation of γ_m with titanium. This is consistent with a local depletion in aluminium across the α_2 platelets determined by fine probe microanalysis. These results suggested that the $\alpha \rightarrow \gamma$ massive transformation occurred via an intermediate fcc phase; and that no orientation relationship or coherency exists between the γ_m and the α grain which was consumed by the massive front.

Substructure of massive transformation product

The observation of APBs in TiAl based alloys that have been water quenched from the α phase field has been studied by a number of investigators; however, there is still no agreement on the mechanism of formation of such boundaries.⁶⁴ According to one investigation, the massive product (γ_m) has an average grain size of about a few micrometres.⁶⁴ Within γ_m grains, a high density of defects caused by the $\alpha \rightarrow \gamma_m$ massive transformation is observed. A large number of fine linear contrast features observed via TEM aligned parallel to one another, lying on the (111) plane, were found to be twins of γ .³⁰

Thermal APBs can be indicative of a primary disordered fcc phase. This hypothesis was discussed

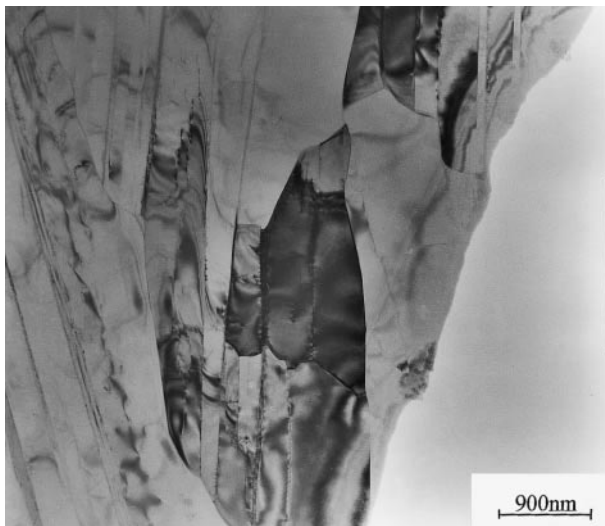
by Jones and Kaufman,²⁹ who questioned the number and scale of variants involved in the observed transformations. They suggested that the APBs were 'grown in' and not thermal in nature. In other words, they proposed that APBs were associated with the $\alpha \rightarrow \gamma$ transformation, rather than formed by the nucleation of ordered domains from a disordered fcc parent phase. To check if it is possible to form a disordered fcc phase, a Ti-48.8Al-2.2V alloy was investigated: under sufficiently large melt under-coolings, γ was the primary phase on solidification rather than the equilibrium α phase, and fully disordered γ was observed.⁹¹ The subsequent sequential ordering process involved nucleation of the three variants of the ordered γ phase and very fine ordered domains were formed.

A later study by TEM showed that the defects that appear as thermal APDBs in γ_m were actually thin 90° domains separating adjacent domains that have the same orientation but are out of phase.^{90,92} This further complicates the process of understanding the origin of these features.

A study of the microstructural features of γ_m showed that γ_m has a large number of faulted domains.⁷⁷ The defects within these domains include dislocations, stacking faults lying on $\{111\}$ planes, APBs, and microtwins. The domains were order domain, twin, or pseudo-twin related, aggregated in colonies that show no specific orientation relationship. The boundaries between these colonies were irregular and do not follow any particular orientation.

Based on the above observations it was suggested that:

- (i) the $\alpha \rightarrow \gamma_m$ transformation corresponds to a massive rather than a martensitic transformation
 - (ii) the nucleation of the γ_m product takes place at grain boundaries, probably by forming a low energy highly coherent interface with at least one of the two matrix grains
 - (iii) the γ_m nuclei have difficulty in growing in the same grain because of the high coherency and tend to grow into the opposite grain, the mechanism for which may be thermally activated jumps of individual atoms across an incoherent α/γ interface
 - (iv) based on the above nucleation and growth process, the apparent absence of a specific orientation relationship between the parent α phase and the γ_m phase was attributed to experimental difficulties
 - (v) two ordering processes were proposed: a direct $\alpha \rightarrow \gamma$ transformation, i.e. order domains and APBs can be formed through the random character of atom transfer, resulting in the arrival of atoms at wrong positions of the lattice; and the existence of a disordered fcc phase followed by an ordering reaction at numerous separate sites and the meeting of two growing domains leading to the formation of ODBs or APBs
 - (vi) twinning was due to a transformation induced stress concentration, which led to the formation of twin and pseudo-twin related domains.
- Several topics are still poorly understood, such as:



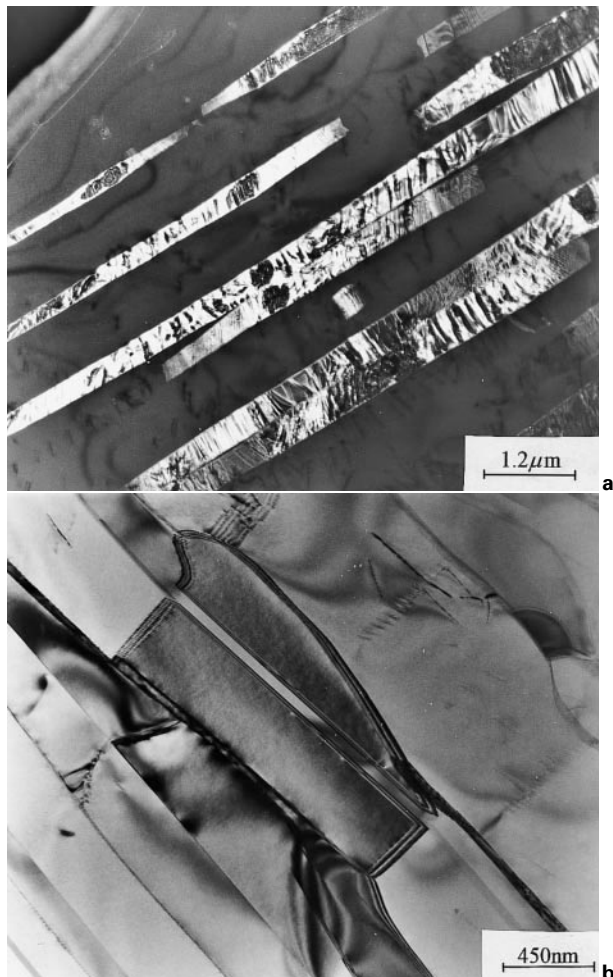
18 Ti-48Al-2Mn-2Nb alloy after heat treatment at 1200°C for 1 week followed by heating to 1420°C for 1 h and water quenching, showing lath colony and adjacent γ grains (TEM)

the origin of the substructure of the massive product, the competition between massive and martensitic products, and the existence of a disordered γ phase. The effect of composition on the nature of the transformation product has yet to be rationalised in thermodynamic terms and information on the nature of nucleation sites is scanty.

Fine duplex microstructure

One of the common types of microstructures obtained on fast cooling, the 'fine duplex' microstructure is formed on water quenching Ti-48Al-2Mn-2Nb alloys from the α phase field.⁹³ This structure consists of equiaxed γ grains and abutting lath colonies (Fig. 18). Both the lath colonies and the grains are composed of the γ phase, twinned γ laths, and α_2 laths. The $\alpha \rightarrow \gamma$ transformation takes place through long range diffusional processes and competitive growth between the lath and equiaxed morphology occurred. Nucleation of the γ phase from the α matrix appears to occur through nucleation on stacking faults followed by growth through sympathetic nucleation and growth of new γ laths on a substrate lath (Fig. 19). The observed misorientations and the interfacial structures between the laths were consistent with such a mechanism. Competition between such nucleation and growth mechanisms for the equiaxed and lath morphologies led to the formation of lath colonies of γ and α_2 interspersed with equiaxed grains.

Similar features were subsequently observed in an investigation of investment castings.⁹⁴ The microstructure of the castings did not show a typical lamellar structure. Instead only γ plates were seen, without any α_2 phase separating the variants. The γ variants did not develop with the commonly observed twin orientations, but were largely related by 120° rotational boundaries. Additionally, tilt and twist boundaries were observed inside γ plates, indicative of the subdomain structure. The interface plane was found to be 3.5° misoriented from the common (111)



a centred dark field image of thin γ phase, showing early stage of γ plate nucleation and growth: note lenticular shape, irregular habit plane, and impingement of plates; b bright field image of individual laths in edge/edge and edge/face configuration, demonstrating sympathetic nucleation and growth mechanism

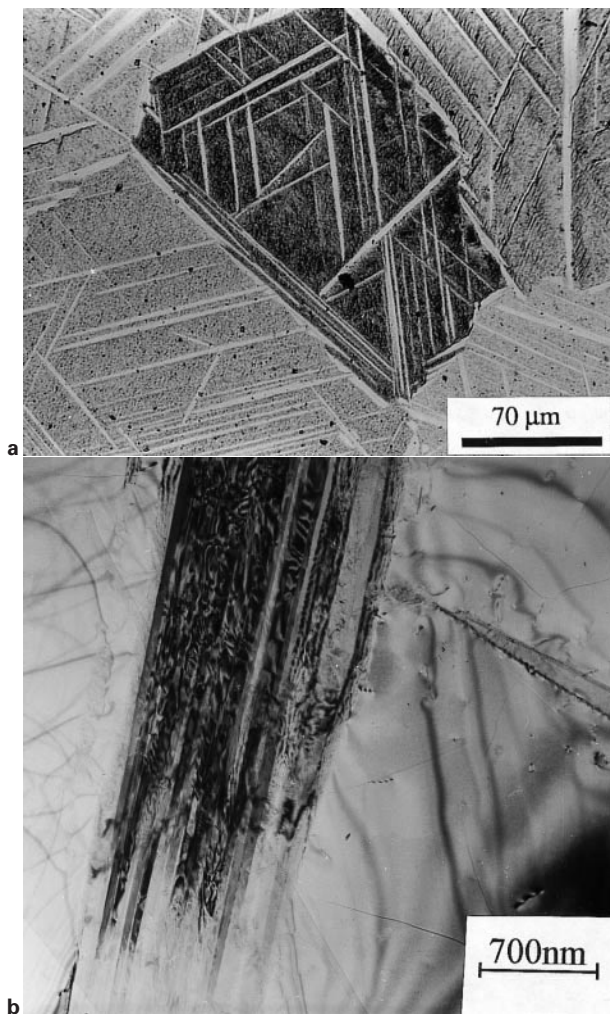
19 γ phase (TEM)

plane. It is clear that the above experimental observations are consistent with a sympathetic nucleation and growth mechanism.

It may be pointed out that an earlier study had suggested an alternative mechanism; fine γ lamellae begin to form on cooling from the α phase field, but the growth rates are slow compared with the heat extraction rates.²⁹ Thus, at lower temperatures, γ branches and grows more rapidly with a less planar interface. However, it has been shown that the crystallographic restrictions on branching preclude this mechanism.⁹³

Transformation in $\alpha + \gamma$ phase field

The precipitation of the α phase from the γ matrix in the $\alpha + \gamma$ phase field was studied.⁹⁵ Alloys consisting of γ grains were heat treated in the $\alpha + \gamma$ phase field and water quenched to room temperature. Precipitation of the α laths in the γ matrix took place at grain boundaries and in the grain interior (Fig. 20a). The Shoji-Nishiyama orientation relationship was observed between the two phases. In the case of binary alloys, the thickening of the α laths



a aged at 1350°C for 30 min followed by water quenching: grain boundary allotriomorphs and thin long laths are observed in the grain interior (optical); *b* aged at 1350°C for 1 h: lath is edge on and consists of fine plates of γ and α_2 (TEM)

20 Ti-50.5Al alloy homogenised at 1200°C for 24 h and aged as indicated

governed the transformation kinetics. A reverse transformation from α to γ occurred within the α lath during the quench and resulted in the formation of fine parallel plates of γ and residual α_2 within the lath (Fig. 20*b*). Disordered α was never observed at room temperature in this composition region, suggesting that the ordering may be a second or higher order phase transformation. The formation of γ within the α lath was found by analytical electron microscopy to be composition invariant. In the case of ternary alloys containing silicon, a change in kinetics was noted with the silicon addition leading to a greater lath density but a smaller thickness, length, and volume fraction than with the binary alloys. The addition of B can completely suppress the transformation of γ to α , but the mechanism has not been understood.

The composition invariant nature of the $\alpha \rightarrow \gamma$ transformation within the α lath was later studied by other workers by analytical electron microscopy.⁹⁶ In Ti-48Al, extremely fine lamellae were formed within the prior α lath. Analysis by EDX with a spot diameter of 5 nm was used to detect composition differences,

and it was concluded that a composition invariant transformation with only short range diffusion to get an ordered structure had taken place. Further investigation of the kinetics and composition changes in this potentially useful, high precipitate density, high interface density microstructure could be rewarding.

Phase transformations in $\alpha_2 + \gamma$ phase field

Effect of temperature

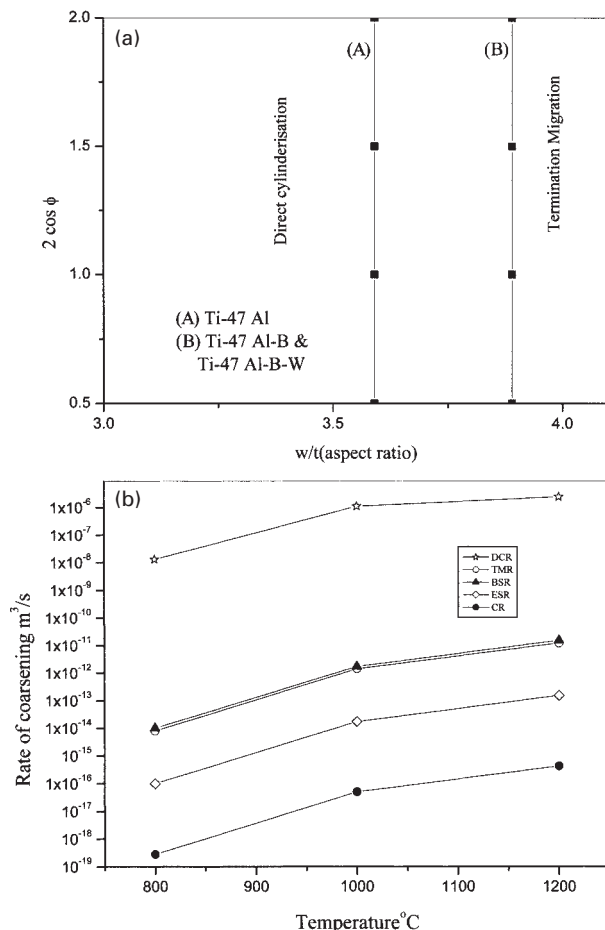
The microstructural stability of these alloys under harsh environmental conditions is of paramount importance in determining the viability of this material for engineering applications. The microstructure degrades when exposed to elevated temperature and stress conditions.⁹⁷⁻¹⁰² A discontinuous coarsening (DC) reaction involving grain boundary migration was observed to occur in as-cast Ti-43Al alloy.¹⁰³ The critical temperature for DC was 900°C. Both regular and 'degenerate' types of secondary lamellae were observed. The interlamellar spacing of the secondary lamellae was >4 times that of the primary lamellae. Growth of secondary lamellae by the edge and sidewise growth mechanisms, similar to that proposed previously for the Fe-C system, was observed. In the initial stages of DC, each grain boundary segment was simply displaced into either grain by edge growth, producing a single seam pattern. As they grow, a double seam pattern of secondary lamellae can develop at grain boundaries because of the sidewise growth of secondary lamellae.

In a study of the morphology and growth kinetics of coarsening of a fully lamellar Ti-44Al alloy,¹⁰⁴ three types of morphologies were observed in discontinuously coarsened lamellar structures. Lamellae were observed with:

- (i) the low energy habit plane as their interface and the same lamellar direction as the primary lamellae
- (ii) the same crystallographic orientation of the α_2 plates, but a different lamellar direction from that of primary lamellae
- (iii) a different crystallographic orientation of the α_2 plates and a different lamellar direction from that of primary lamellae, but where the low energy habit planes comprise the lamellar interfaces.

At the macroscopic level, a study of coarsening of the lamellar structure in two phase Ti-47.5Al indicated that coarsening was induced by nucleation and growth of hole-like macrodefects. Comparison of the predicted kinetics of coarsening with experimental data was also performed.¹⁰⁵

The thermal stability after aging at elevated temperatures has been studied by TEM and AEM.^{98,99} Predominantly lamellar microstructures of Ti-47Al and Ti-47Al-0.5W and Ti-47Al with 140 at.-ppm B additions were heat treated at 800, 1000, and 1200°C for 168 h. Aging at 800°C did not show much change, but aging at 1000°C produced significant lamellar breakup and coarsening in the binary and ternary alloys, although not in the quaternary alloy. Discontinuous coarsening was also observed, leading



21 a instability maps showing dominant instability mechanisms for three alloy compositions aged at 1200°C for 1 week; **b** variation of coarsening rate with temperature for unit volume of lamellar structure for Ti-47Al-0.5W alloy with 40 at.-ppm B addition: DCR = discontinuous coarsening rate, TMR = termination migration rate, BSR = boundary splitting rate, ESR = edge spheroidisation rate, CR = cylinderisation rate

to the formation of irregular, coarser, and branched lamellae. Aging at 1200°C led to significant degradation of the microstructure and, in addition to the instabilities observed at lower temperatures, spheroidised precipitates of α and γ were observed. Cut surfaces showed additional features not observed in the bulk, such as loss of aluminium from the sample and the recrystallisation and growth of new grains into the lamellar structure. After aging at 800 and 1000°C such subsurface instabilities were greatest for the binary alloy and least for the quaternary alloy.

Instability maps can be constructed for such changes in the microstructure.¹⁰⁰ Such maps have been developed for an isolated plate based on experimental data^{98,99} (Fig. 21). These maps showed that the dominant mechanism controlling the coarsening changed with time and that the coarsening of lamellae (with an initial large aspect ratio) was controlled by termination migration. Below a critical aspect ratio, cylinderisation dominates. Instability maps were also constructed using the criterion of maximum mass

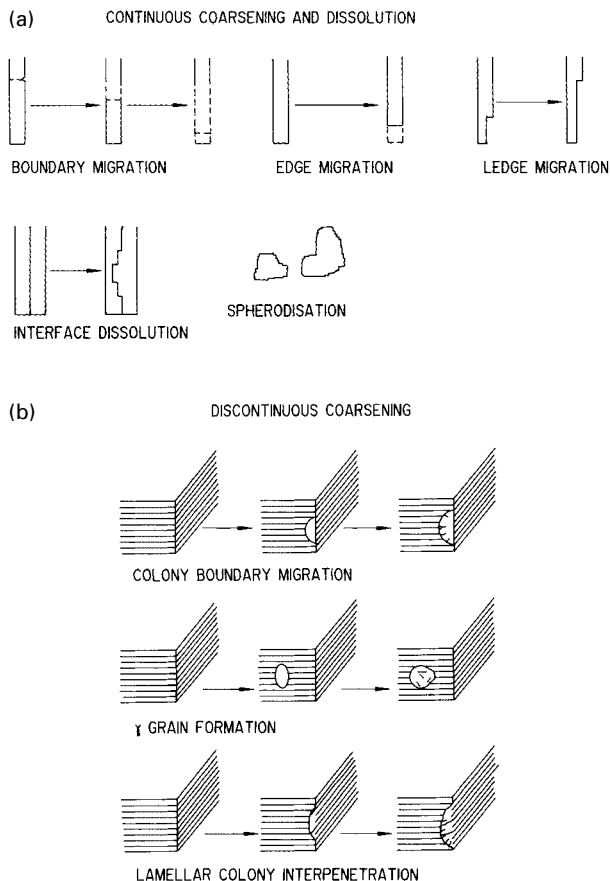
transport per unit volume. Such maps showed that the coarsening rate associated with unit volume of the lamellar structure depended on the density of faults and that termination migration and boundary splitting occurred at a considerable rate compared with other continuous coarsening (CC) mechanisms. The rate of DC was found to be high. The effect of boron and tungsten was also shown to reduce the coarsening rate of the initial lamellar structure.

The growth kinetics of DC of the lamellar structure in Ti-44Al-0.5Cr alloys was studied by optical microscopy and TEM.¹⁰⁶ The growth kinetics in the range 923–1473 K was measured to be linear in the initial period of the reaction and reached a maximum near the eutectoid temperature. The migrating grain boundary plane maintained no definite geometrical angular relationship to the lamellar interface and DC occurred by grain boundary diffusion. This kinetic analysis suggested that the driving force for grain boundary migration was the elastic strain energy rather than the chemical and interfacial free energies, despite the comparatively large value of the last two factors. The reasons for this were not clearly understood. The possibility of dissipation of the chemical free energy in front of the migrating grain boundary was evaluated.

Aging of a γ based alloy showed that when the heat treatment temperature in the $\alpha + \gamma$ phase field increased, DC became less pronounced and tended to disappear.⁷⁷ In as cast Ti-46Al alloys heat treated at 1300°C, the γ grains started from the initial interdendritic regions and/or at grain boundaries. The disappearance of DC and occurrence of CC with increasing temperature has been observed in many alloys. The stability of the lamellar structure was promoted by low energy γ/α_2 interfaces. Discontinuous coarsening could be divided into two processes. First, formation of secondary γ lamellae almost parallel to the primary γ lamellae, behind the migrating α or α_2 grain boundaries. The secondary lamellae were much coarser and the boundaries between the α_2 and γ phases were less regular, although the Shoji-Nishiyama orientation relationship was maintained. The secondary lamellae were single variant and do not show any ODBs. Second, optical microscopy showed DC on both sides of a grain boundary (double seam development). The presence of serrated grain boundaries may be related to the early stages of DC or a discontinuous precipitation reaction.

Effect of stress

The effect of 5.5% cold forging followed by aging at 1000°C was studied in Ti-47Al and in Ti-47Al and Ti-47Al-0.5W containing 140 at.-ppm B to understand the behaviour of the cut surfaces and to study the effect of prior cold working.^{98,99} Prior cold working was found to cause considerable recrystallisation in the binary alloy but almost none in the quaternary alloy. The following defects were found to be mainly responsible for the instabilities: low angle grain boundaries, misoriented lamellae, lamellar interfaces, and colony boundaries. The observed modes of instability were fault and edge migration,



a related to diffusional processes in grain interior; *b* related to grain boundary diffusional processes

22 Schematic illustration of mechanisms of instability observed in γ based titanium aluminides

the dissolution of lamellae at lamellar interfaces, and the formation of new γ grains within the lamellar structure (Fig. 22a). At 1200°C, spheroidisation was also observed. Grain boundary related instabilities included the migration of colony boundaries, the penetration of one colony into an adjacent colony, and the growth of γ grains into adjacent lamellar colonies (Fig. 22b). Boron and tungsten refine the initial lamellar structure and make it more resistant to coarsening (and recrystallisation if cold deformed) during aging. These effects were related to the fact that boron refines the structure relative to the binary alloy but leads to the formation of fragmented and discontinuous α_2 lamellae. Boron and tungsten together refine the structure and produce more uniform and continuous α_2 lamellae.

In a study of creep of a γ based alloy with a lamellar structure it was shown that at temperatures between 1000 and 1050°C, the dominant coarsening mechanism was intrapacket termination migration.^{107–109} After aging for 2710 h at 1050°C, a coarse globular structure was formed. Other coarsening mechanisms which contributed in a minor fashion were cylinderisation, Ostwald ripening, interpacket termination migration, and DC. The effect of thermo-mechanical exposure was significantly to increase secondary creep rate because of severe distortion (bending and kinking) and complete transformation



23 Dynamic precipitation at γ /lamellar boundaries (TEM)

to a spheroidised structure. Creep caused progressive deformation induced spheroidisation, which occurred by deformation and fragmentation of lamellae in localised shear zones at interpacket boundaries and within lamellar packets. Deformation induced spheroidisation is an alternate mechanism to capillarity induced spheroidisation.

An investigation of creep deformation and creep microstructure of a Ti–48Al–2Cr alloy was conducted¹¹⁰ and the loss of creep strength during tertiary creep was found to be associated with dynamic recrystallisation of single phase γ regions and spheroidisation of α_2 laths.

Severe microstructural changes were observed in duplex material following creep.¹ These changes included precipitation of interstitial impurities, changes of the interfacial boundary structure, dynamic recrystallisation of equiaxed γ grains, and the spheroidisation of the α_2 laths. The γ lamellae next to the dynamically recrystallised γ grains showed some buckling and also experienced dynamic recrystallisation at larger creep strains.

At the TEM level of observation, the role of ledges in creep of TiAl alloys with fine lamellar structure has been investigated.¹¹¹ Creep experiments have been conducted on two alloys prepared by powder metallurgical processes, with fine grained, fine lamellar structures, of composition Ti–47Al–2Cr–2Nb and Ti–47Al–2Cr–1Nb–1Ta. Thinning and dissolution of α_2 lamellae and continuous coarsening of γ lamellae were observed at both low and high stresses, with cross-twinning operating only at high stresses. A ledge creep mechanism involving branching and escaping of unit height ledges into or from multiple height ledges at interfacial boundaries was proposed. The activation energy for creep in the high stress regime was close to the energies for lattice self-diffusion of titanium in γ and α_2 and for interdiffusion in γ and α_2 .

Dynamic effects

Dynamic effects including interphase precipitation occur at migrating interphase interfaces. Interphase precipitation has been observed during long term isothermal annealing and under creep conditions¹¹² (Fig. 23). Such precipitation is usually observed in

C–Cu–Ni steels and vanadium alloyed carbon steels. Titanium aluminides can also show this type of precipitation, which is otherwise rare in titanium base alloys. A Ti–47Al–0.5W alloy containing 140 at.-ppm B was aged at 1000°C for 1 week; TEM indicated that γ grains grew into the initial lamellar structure. Such growth was accompanied by precipitation at the γ grain/lamellar colony interface. These precipitates were found to belong to the α_2 phase or the $\text{Ti}_2\text{Al}(\text{C,N})$ phase. Their morphology was unusual, being faceted with an aspect ratio of approximately 2:1 and bounded by (0001), (10 $\bar{1}$ 0), and (11 $\bar{2}$ 0) type habit planes. The Shoji–Nishiyama relationship has been established between the growing γ grain and both the α_2 and the $\text{Ti}_2\text{Al}(\text{C,N})$ precipitate.

The effect of heat treatment on precipitation and interphase precipitation has been studied in a Ti–47Al–2Nb–1Mn–0.5W–0.5Mo–0.2Si alloy.¹¹³ After a two step heat treatment, interphase precipitation was observed in grain interiors and interfaces and improved the creep resistance. The heat treatment caused precipitation and the growth of the precipitates was affected by the stress state, such that the stressed region of the material stimulated more homogeneous and faster precipitate growth. Moving dislocations, acting as pipes for transport of solute atoms to precipitates, enhanced and homogenised precipitate growth. This effect is beneficial for a creep resistant structure that does not develop larger, harder inclusions that could degrade ductility. A Ti–47Al–2Cr–2Nb alloy also developed fine precipitates during a two step heat treatment, but the substantial improvement in creep resistance was also correlated with a much larger grain size.

Thermomechanically processed structures

General effects

Microstructural changes that occur during hot working are of immense importance in determining the final grain size, the lamellar spacing, and the range of workability of γ based alloys. Hot extrusion, pack rolling, and conventional hot forging methods have been used for hot working,^{114–117} which is often used to break up the initial inhomogeneous cast structure in large ingots. The microstructure in heat treated cast material can be generated in wrought material at a much more refined level and with considerably improved homogeneity. Wrought processing followed by heat treatment is often the basis for processing.

Cast processing often does not give optimum mechanical properties because of limited microstructural control.²⁶ On the other hand, TMP increases the spectrum of microstructures and a better balance of properties can be obtained. The TMP route is used especially to homogenise large ingots. Fine grained fully lamellar structures having controlled lamellar characteristics can be obtained. Elements favouring β phase formation (e.g. 1 at.-%Mo) and boron can be used as grain size controlling agents. Such elements also significantly change the lamellar transformation kinetics, thus

changing the thermal process path and affecting the perfection of the lamellar structure.

In the ingot metallurgy route, cast ingots are vacuum or plasma arc melted, hipped, and then hot worked in the $\alpha + \gamma$ phase field by isothermal forging, hot die forging, extrusion, or rolling. To refine the initial coarse and inhomogeneous cast structure, a two step hot working process is generally adopted.¹¹⁴ Owing to the high strain rate sensitivity of the flow stress, the hot working parameters (temperature, strain rate, extrusion ratio, canning technology, etc.) must be optimised carefully. Two step isothermal forging of cast ingots in the $\alpha + \gamma$ phase field leads to recrystallised uniform fine grained microstructures. Heat treatment of the hot worked structure near the eutectoid temperature produces the ‘near γ ’ structure consisting of equiaxed γ grains and coarse α_2 precipitates. The uniformity of the equiaxed γ grain size depends on the hot working conditions and is increased by multistep processing.

Primary processing of ingots is typically by wrought processing. Secondary TMP is used to further refine the grain size and improve the homogeneity of primary wrought structures, control their textures, and produce mill product shapes. Such processing usually begins with annealing of the products from primary ingot breakdown. This annealing at a temperature approximately midway within the $\alpha + \gamma$ phase field promotes recrystallisation of γ and lamellar morphologies as well as spheroidisation of the α phase (and β phase in three phase alloys). These annealed structures may then be forged to pancakes, extruded, or rolled into sheets, either isothermally or by canning. The working temperature, strain, strain rate, and deformation axis can be chosen so as to decrease the grain size to submicrometre levels; thus superplastic microstructures are readily formed.

Microstructural changes in production scalable processes

Three production scalable processes have been studied: isothermal forging, conventional (canned) forging, and conventional (canned) extrusion.^{118,119} For each approach, the ingot was hipped or given a homogenisation heat treatment before working. Hipping was carried out at about 1260°C at a pressure of about 175 MPa. In isothermal forging, the lamellar structure spheroidises typically to about 50%. The kinetics of globularisation of Ti–45.5Al–2Cr–2Nb were examined and it was found that spheroidisation initiated and proceeded inward from prior α grain boundaries. The grain interiors showed moderate to extensive kinking of the lamellae, depending on the orientation of the lamellae relative to the applied load with respect to the breakdown of the lamellar structure; and spheroidisation kinetics that increased as the strain rate decreased, for a given α grain size, and decreased with increasing α grain size for a given strain rate. The dependence on the percentage of spheroidisation on the strain followed a sigmoidal behaviour.

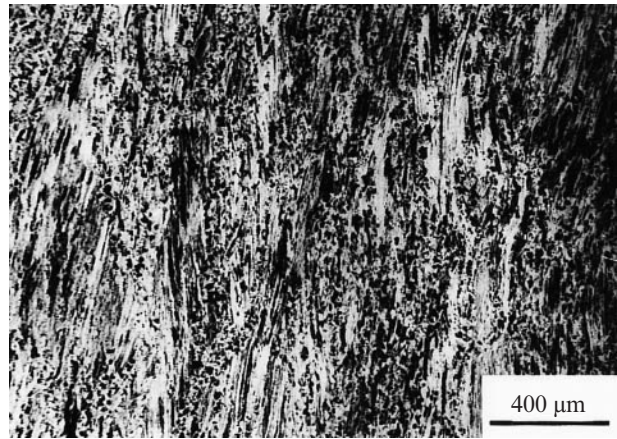
The as forged microstructure produced by conventional hot forging was typically finer than that from isothermal forging, with very few remanent lamellar

colonies. With optimal can design, insulation, temperature, deformation non-uniformities can be minimised during conventional hot forging and relatively uniform macro- and microstructures can be obtained. In the case of canned hot extrusion, a grain size variation was usually observed. Using equichannel angular extrusion, this microstructural variation can be minimised.

The development of uniform, fine microstructure during primary processing is useful in secondary processing (sheet rolling, superplastic forming, isothermal forging). The fine equiaxed γ grain microstructure that enables superplastic forming or isothermal forging has inferior mechanical properties compared with a fully lamellar structure with moderately small α grains. Processing techniques to obtain the latter include supertransus heat treatment of alloys containing grain growth inhibiting elements such as boron, in solid solution or in the form of precipitates; and controlled transient heating in the α phase field.

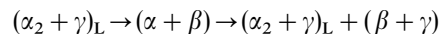
The microstructural development during conventional and isothermal hot forging of a γ based alloy was studied.¹¹⁸ The breakdown of the lamellar structure in an ingot melted Ti-45.5Al-2Cr-2Nb alloy was investigated under canned, conventional hot forging and isothermal forging. The higher rate of deformation in conventional forging led to considerably finer and almost completely broken down structures in the as forged condition. Several non-traditional approaches, including isothermal forging of a metastable microstructure (α forging), and the inclusion of a short static recrystallisation anneal during forging, produced a more fully broken down structure than in an isothermally forged billet. Despite the microstructural differences after forging, conventionally and isothermally forged material responded similarly during heat treatment. In both cases, almost completely recrystallised structures of either equiaxed γ grains or transformed α grains surrounded by fine γ grains were produced depending on the heat treatment temperature. It was concluded that deformation processes often produce partially recrystallised microstructures in cast or cast and hiped ingot materials with a lamellar microstructure. Stored energy from such processes produced more extensive static recrystallisation and globularisation. However, the uniformity of microstructure on wrought and recrystallisation annealed materials was influenced by the presence of prior interdendritic, single phase γ regions produced during ingot solidification. Such regions occurred as a result of the double peritectic reactions. The kinetics of homogenisation heat treatment to remove such non-uniformities have been established, since they result in unwanted banded microstructures in wrought products.

The microstructural evolution needed to obtain an intermediate colony size in a fully lamellar structure has been studied.¹²⁰ A series of forging and post-forging heat treatments have been conducted on plasma arc melted ingots of a Ti-46.5Al-2Cr-2Nb-0.8Mo-0.2W-0.2Si alloy. Grain size refinement and dynamic recrystallisation were observed. Heat treatment following forging yielded nearly fully lamellar structures while minimising grain growth.



24 Fine recrystallised grains and remnants of lamellar structure in hot worked material (optical)

Appropriate variable choice yielded lamellar colonies surrounded by coarse intergranular $\beta + \gamma$ phases. Forging at temperatures above T_x began with large lamellar colonies breaking down into α grains and β particles. The β particles stabilise the α grain size. Lamellar colonies and intercolony $\beta + \gamma$ structures resulted during cooling from the forging temperature. This process can be summarised as

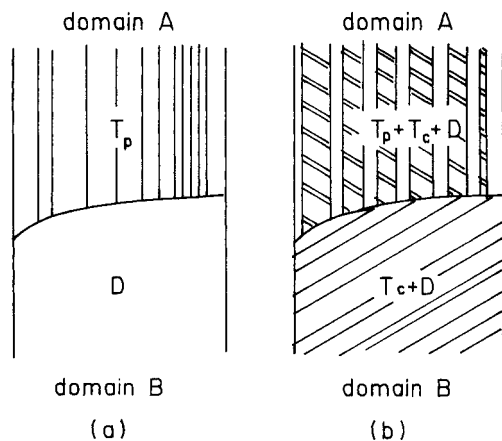


where the subscript L denotes the lamellar morphology. This $\beta + \gamma$ region at the prior α grain boundary slowed down grain boundary growth, producing relatively small lamellar colony sizes. The cooling rate from the heat treatment temperature was crucial in obtaining lamellar structures, with too rapid cooling giving Widmanstätten or acicular α_2/γ colonies and too slow cooling giving rise to a large $\beta + \gamma$ volume fraction.

Examples of microstructural changes during hot working

1. A TEM study of the non-equilibrium α_2/γ interfacial structure in a Ti-45Al-10Nb-2.5Mn-0.05B alloy after hot working showed that non-equilibrium interfaces with a misorientation from the usual orientation relationship have been observed after annealing.¹²¹ The structure of these interfaces was that of numerous interfacial ledges containing $\frac{1}{3}[111]$ Frank partial dislocations. These interfaces may be formed by the relative rotation of α_2 and γ plates in the lamellar structure. Many low angle grain boundaries were formed in the γ plates, necking of α_2 plates occurred preferentially at α_2 subgrain boundaries, and finally spheroidised α_2 particles were observed.

2. Microstructural changes of a Ti-46Al alloy during multistep working were studied.¹²² Multistep (1 to 10 steps) compression tests were conducted at 1100°C at strain rates of about 10^{-3} s^{-1} and a total reduction of 80%. The lamellar structure was destroyed and the morphology changed to a dual phase fine ($\sim 10 \mu\text{m}$) grain structure of γ and α_2 grains. After 10 steps the lamellar structure was 95% spheroidised.



a cold rolled <20%; b cold rolled >40%

25 Schematic illustration of deformation structures in cold rolled PST crystals: D, T_p , and T_c represent dislocations, deformation twins parallel to the lamellar boundaries, and twins lying across lamellae respectively

3. Hot die forging of Ti–48Al and Ti–48Al–2Mn–2Nb alloys has been studied.¹¹⁵ Sound forgings could be achieved in such cases using high billet temperatures (1250–1350°C) and high die temperatures (1000°C). As a result of the inhomogeneous cast microstructure the initially circular cross-section of the forgings became elliptical. Hot die forging led to non-uniform macrostructure (formation of dead zones) and non-uniform microstructure, consisting of a network of lamellae and fine recrystallised γ grains with a size varying from 2 to 20 μm (Fig. 24). The volume of recrystallised grains depended on both billet and die temperatures. Large scale recrystallisation typically observed in isothermal forging was restricted by the high strain rates and short times used in hot die forging. Increasing the die and billet temperatures decreased the non-uniformities.

4. Deformation and recrystallisation of lamellae was studied using PST crystals.^{123–125} Isothermal hot working can produce a fine microstructure. When the lamellar boundaries lie at 45° from the rolling direction, cold working to about 50% reduction in thickness was possible. When the reduction in thickness was less than 20%, γ lamellae were deformed by shear parallel to the lamellar boundaries. Since γ lamellae were composed of six different types of domain, some domains deformed by slip and others by twinning (Fig. 25a). As the reduction was increased, shear deformation via both slip and twinning across the γ lamellae became increasingly dominant (Fig. 25b).

Alloying effects

The role of alloying effects on the phase transformations has been much less studied than their effect on mechanical properties.^{126–128} Alloy compositions are typically in the range 45–52%Ti, 45–48%Al, 1–3%(Cr,Mn,V), and 2–4%(Nb,Ta,Mo,W). A study combining alloying effects and deformation processes in duplex microstructures attempted to increase the ductility by promoting slip transfer across α_2/γ interfaces.¹²⁹ Slip transfer would enable the transfer

of plastic deformation across the α_2 lamellae so that deformation twins could propagate continuously from one γ lamellae to the next. This can be done by balancing the ease of deformation in the two phases, i.e. by strengthening the γ phase and softening the α_2 phase. Zirconium and niobium were shown to be possible γ phase strengtheners, while chromium is a possible α_2 softener.

No single alloying composition has emerged as the optimum; however, aluminium contents are generally between 45 and 48%, niobium is added to increase the oxidation and corrosion resistance, and chromium and manganese to increase the ductility. Vanadium, manganese, and chromium do not ductilise single phase γ or fully lamellar structures, only the duplex structure.¹²⁶ Oxygen increases both strength and creep resistance.

The effect of alloying elements on mechanical properties¹ can be summarised as follows. A decrease in aluminium content increases strength and decreases ductility and oxidation resistance. In binary alloys, decreasing aluminium content below 48% decreases toughness and creep resistance. Vanadium, manganese, molybdenum, and chromium increase the ductility in two phase alloys. At room temperature, molybdenum, tantalum, niobium, vanadium, chromium, and manganese increase the strength of the γ phase. Solid solution hardening is effected by addition of vanadium, chromium, manganese, molybdenum, tantalum, niobium, and zirconium, of which the last four are the most effective. The refractory elements tantalum, molybdenum and tungsten, as well as carbon, increased high temperature strength retention. Carbon increased the creep strength if an appropriate carbide dispersion was present. Additions of carbon also decreased creep rate. The addition >3% of solute may result in β or B2 phase formation.

Boron and silicon can refine the duplex microstructure, and may also be added for grain refinement and to stabilise the microstructure. Silicon increases creep resistance. Boron can restrict the growth of α grains when alloys are heat treated in the temperature range: $T_\alpha - 10$ to $T_\alpha + 20$ K, where T_α is the α transus temperature, thus generating nearly 100% lamellar or 100% (fully) lamellar structures. Fully lamellar structures lead to better creep strength and fracture toughness, but decreased ductility compared with duplex microstructures.

Precipitation hardening

The low solubility of oxygen and nitrogen in the γ phase (the maximum solubility of oxygen is about 250 ppm) causes oxides to form; similarly, carbides and nitrides can also form needle shaped $\text{Ti}_3\text{Al}(\text{C,N})$ precipitates with the perovskite structure. Hardening by fine dispersions of such precipitates may be useful to increase the high temperature strength and creep resistance. Silicon can also be used to form Ti_5Si_3 precipitates.

A summary of the effects of various alloying elements is presented in Table 1.¹²⁶

Selected studies

Niobium, tantalum, tungsten, and molybdenum have been found to substitute for titanium to produce a

system having a phase diagram in which the solubility limit of aluminium in the γ phase is increased.⁷ Vanadium, manganese, chromium, and silicon can substitute for both titanium and aluminium to give a phase diagram in which the γ aluminium concentration is unchanged or mainly replace aluminium to decrease this concentration.

Boron effects

The microstructural evolution in alloys containing β stabilisers and boron additions was studied in Ti-47Al-2Cr-2Nb-1Mo-(0-0.2)B alloys.¹³⁰ The sequence of transformations was: $L \rightarrow L + \beta \rightarrow \alpha + \beta$ cores $\rightarrow (\alpha + \gamma)_L + \beta$ cores $\rightarrow (\alpha_2 + \gamma)_L + B2$ cores. Boron restricted α grain growth, promoted the lamellar γ morphology, and increased the lamellar growth rate. Boron and TiB_2 are used as inoculants in as cast alloys.^{3,10} Some γ based alloys have either boron or TiB_2 additions, which reduce the grain size.¹³¹ Approximately 0.8 vol.-% TiB_2 is required to decrease the grain size. Fine grained fully lamellar structures can be formed by heat treating boron containing alloys (so called TMTL structures).²⁶ The boron contents are in the range 0.03–0.5 at.-%. Boron at this level is ineffective as a grain refiner for castings, but assists grain size control during solid state processing of wrought alloys. The boride morphology in a γ based alloy containing iron, vanadium, and boron showed that in an alloy containing the B2 phase, the TiB_2 morphology has the form of very high aspect ratio plates interleaved with B2 on a nanoscale in flakelike dispersoids.¹³² The TiB_2 and B2 interfacial relationship was $\{10\bar{1}0\}_{TiB_2} \parallel \{001\}_{B2}$ and the grains grow preferentially in these planes, producing dispersoids with a $\{10\bar{1}0\}_{TiB_2}$ habit plane. Within each dispersoid the orientation relationship was $\{10\bar{1}0\}_{TiB_2} \parallel \{001\}_{B2}$, $\langle 0001 \rangle_{TiB_2} \parallel \langle 010 \rangle_{B2}$. Only EELS spectra taken from the dispersoid revealed boron edges, indicating that there was strong segregation of boron to the dispersoid regions, rather than to one of the bulk phases or to interfaces.¹³²

Table 1 Effect of selected alloying elements in γ based alloys¹²⁶

Element	Effect
Nb	Increases oxidation resistance and creep resistance
P	Doping with P decreases oxidation rate
Si	Addition in 0.5–1% range increases creep resistance. Silicon also increases oxidation resistance and fluidity and decreases susceptibility to hot cracking
Ta	Increases oxidation and creep resistance and tendency for hot cracking
V	Addition in 1–3% range increases ductility in duplex alloys. Reduces oxidation resistance.
W	Oxidation and creep resistance increased
Al	Strongly affects ductility by changing microstructure, best ductility in range 46–50% (increasing Al in this range decreases fracture toughness)
B	Additions >0.5% refine grain size, increase strength, workability, castability
C	Increases creep resistance and decreases ductility
Cr	Additions of 1–3% increase ductility of duplex alloys, additions >2% increase workability and superplasticity, additions >8% increase oxidation resistance
Mn	Additions of 1–3% increase ductility of duplex alloys
Mo	Increases ductility, strength, and oxidation resistance

Creep reducing elements

The effect of tungsten in solution is to decrease creep rate.¹³³ Tungsten may hinder the glide of dislocations, or as a result of the low diffusivity of tungsten, reduce the recovery rate. Niobium and tantalum decrease creep rate by decreasing diffusion rate,¹⁰ and increase the creep strength and environmental resistance without unduly decreasing ductility. Nitrogen and carbon may provide dislocation drag; 0.2–0.3% C additions can lower the creep rate.¹³⁵ Tungsten and silicon increase creep resistance through microstructural modification. Oxygen increases the strength, but may or may not decrease the ductility, depending on the oxygen concentration and whether α_2 was present. Tungsten, tantalum, and molybdenum are solid solution strengtheners and decrease diffusion rates. The β phase decreases creep resistance.

Hydrogen

Little hydrogen is absorbed at near atmospheric pressures.¹³⁴ Two hydrides can form under high pressure, but the solubility limits are unknown. Considerable strengthening accompanies charging and the fracture mode is brittle and complex.

Summary

The development of γ based titanium aluminides for high temperature load bearing applications has been shown to be dictated by a deep understanding of the phase transformations and microstructure development in this system. The microstructures produced by different processing techniques and by changing the thermomechanical processing conditions have been discussed. The lamellar microstructure and the formation of twins has been the object of study by many investigators and the proposed mechanisms of formation are outlined. The complex effect of cooling rate on the morphology is discussed and the effect of other heat treatment variables has been delineated. Microstructural stability developed as a result of heat treatment, cold work, and hot working has been shown to be amenable to improvement by, for example, alloying additions. Dynamic precipitation, a relatively unusual phase transformation in titanium based alloys, is also presented. It is clear that several routes exist to use the current knowledge of phase transformations to improve the properties and thus the performance of this important class of intermetallic compounds.

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